Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Tests

Daniel Zhao

Supervisor: Associate Professor Joseph T. Lizier

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School of Computer Science
The University of Sydney
Australia

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Abstract

Mutual information (MI) measures how much observing one variable reduces uncertainty about another. This thesis considers whether MI between two categorical variables differs across independent populations. A Normal Wald test provides one approach, but small or sparse samples can distort the estimated difference and its standard error, producing unreliable false-positive rates and power.

This thesis develops an Expanded Welch–Satterthwaite test that retains Wald's estimated difference and standard error and uses a Student t distribution. Its degrees of freedom depend on how much the estimated MI variance changes between samples. The thesis derives a first-order approximation to that variability by accounting for changes in cell probabilities, their margins and the resulting MI values.

Simulations check the moment approximations and compare false-positive rates and power with Normal Wald. The mean and sampling-variance approximations have different errors, and estimated degrees of freedom can remain well below the target based on the actual sampling moments. An online-shopping dataset illustrates interpretation on observed data.

Expanded Welch brings false-positive rates closer to the intended level in selected small-sample populations at common MI 0.02 nats, at a cost in nominal power. Its power advantage after empirical calibration depends on which equal-MI population supplies the cutoff and reverses in some comparisons when that population changes. A larger cutoff also leaves sparse-table bias and inaccurate standard errors unresolved. The results support selected calibration improvements while identifying estimation errors that limit reliable inference.

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Chapter 1

Introduction

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another (Bossomaier et al., 2016). For example, a categorical input feature X may carry different amounts of information about a class label Y in two populations. This thesis develops and evaluates a statistical test for that comparison using two independently collected samples.

Let P and Q be the joint probability distributions of X and Y in the two populations. The null and alternative hypotheses are

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q). \tag{1.1}$$

Under H_0, the populations have equal MI. However, this does not mean that their joint probability tables are the same, as two populations can have equal MI while having different probability tables.

In practice, MI is estimated from observed data. Two samples can produce different MI estimates even when the population MI values are equal, because their observed counts vary from sample to sample. A statistical test must account for this sampling variation before an estimated difference can be interpreted as evidence of a population difference.

1.1 Why the comparison is difficult

The Normal Wald test (Wald, 1943; van der Vaart, 1998) divides the estimated MI difference by its standard error and compares the resulting statistic with a standard normal distribution. At a 5% significance level, the intended false-positive rate is 5% when the population MI values are equal. The normal approximation can be

inaccurate in small or sparse samples, causing the test to reject equal MI too often or too rarely.

Both the MI difference and its standard error are estimated from count tables, whose cells record how often each pair of categories occurs. MI depends on the cell probabilities and their row and column totals. When samples are small and rare combinations of categories occur only a few times or not at all, the resulting probability estimates can distort both the MI difference and its estimated uncertainty.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but uses a Student t distribution instead of the standard normal distribution. Choosing its degrees of freedom requires estimating how much the MI variance estimates change from sample to sample. For MI, this calculation must account for the fact that changing a cell probability also changes its row and column totals and the MI values calculated from them. This thesis derives how these probability changes affect the estimated MI variance and uses that calculation to obtain degrees of freedom for Expanded Welch.

The Student t distribution gives Expanded Welch a larger rejection cutoff than Normal Wald, so it rejects less often. This can reduce false positives, but it can also reduce statistical power: the probability of rejecting equal MI when the population MI values genuinely differ. The evaluation measures the resulting false-positive rates and nominal power, then compares power after matching rejection at specified equal-MI populations through independent simulation. Changing the calibration population checks whether the resulting power comparison is stable. The frequency of usable p-values is reported alongside these rates.

1.2 Research questions

The thesis addresses two research questions:

1. How accurately do the derived approximations predict the mean and sampling variance of the estimated MI variance, and what degrees of freedom do they imply near and away from independence?
2. How do Expanded Welch's false-positive rates and power compare with Normal Wald?

The simulations first check the variance approximations, then compare false positives and power. Power is measured both at the stated 5% cutoffs and after independent adjustment to approximately equal false-positive rates.

1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance, including the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this variance sensitivity to choose the degrees of freedom for a two-sample test using a Student t distribution.
3. **Empirical contribution.** Controlled simulations check the variance approximations and compare false-positive rates, power and usable results as sample size, table size, category frequencies and MI change. Further comparisons examine association pattern and sample allocation.
4. **Practical contribution.** The evaluation identifies selected populations at common MI 0.02 nats where Expanded Welch brings false-positive rates closer to 5% than Normal Wald, the dependence of calibrated power on the fitting population, and estimation errors that make both tests unreliable.

1.4 Scope and organisation

The scope of this thesis is the comparison of MI between two populations using categorical data. Each sample is summarised in a count table, and both tables use the same fixed row and column categories. The two samples are independent, and observations within each sample are also independent. For the main derivation, both populations are assumed to have positive MI and a positive probability in every cell. The case of zero MI, where X and Y are independent, is considered separately. The main simulations compare Normal Wald and Expanded Welch using two-sided tests at the 5% significance level. An independently fitted moment reference helps examine the role of the degrees of freedom.

Chapter 2 reviews the background and literature. Chapters 3 and 4 define the statistic and derive Expanded Welch. Chapters 5 and 6 present the experiment, results and online-shopping application. Chapters 7 and 8 discuss the implications and limitations. The appendices give supporting calculations, experimental settings and additional results.

Chapter 2

Background and Literature Review

Testing for equal MI in two populations requires an estimate from each sample and its sampling variance. This chapter develops the MI estimator, derives its variance and reviews tests for equal population MI.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \tag{2.1}$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y), the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y.

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. The numbers of categories, r and c, are their alphabet sizes. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the marginal probabilities, or row and column margins. The pointwise mutual information (also called local MI) associated with cell (i, j) is

$$\ell_P(i, j) = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts, whereas a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r}\sum_{j=1}^{c} p_{ij}\ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row i, the inner sum runs over all columns j. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y), we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

MI is therefore the average reduction in uncertainty about X obtained by observing Y, or equivalently about Y obtained by observing X. It is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. The null hypothesis requires equal MI, not identical cell probabilities in the two population tables.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij}. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

The plug-in estimator inserts these observed relative frequencies into Equation (2.4):

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x\downarrow 0} x \log x = 0$, so empty cells contribute no term to the sum. When a positive-probability cell is unobserved, the empirical table assigns it zero probability, contributing to sparse-sample estimation error.

Under multinomial sampling, the cell counts vary between samples but must sum to n. An extra observation in one cell therefore leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its finite-sample bias and sampling variance depend on the complete probability table. In particular, rare cells can change both even when the categories and sample size are unchanged (Paninski, 2003).

For a fixed positive $r \times c$ support, the large-sample expansion of the plug-in estimator's bias has a positive leading term:

$$\text{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The leading term grows with table size and shrinks as the sample grows, although it does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing MI estimates from two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2}\left(\frac{1}{n_P} - \frac{1}{n_Q}\right). \tag{2.9}$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MI values. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

Estimation beyond the leading bias correction.

Data are undersampled when too few observations are available to estimate the joint probabilities reliably. Rare category combinations can then be poorly represented or absent from the sample. Panzeri and Treves (1996) derive analytical bias corrections that account for the categories with nonzero population probability, but these remain large-sample approximations. Paninski (2003) shows that substantial error can remain after standard corrections, including the jackknife, when many categories are poorly sampled. Chao and Shen (2003) instead adjust entropy for categories missing from the sample. Their method uses sample coverage, the total population probability of the categories that have been observed, and accounts for the different probabilities of discovering common and rare categories.

Bayesian estimators model the unknown probabilities rather than treating observed proportions as the complete population. Nemenman et al. (2002) combine priors, or probability weights on possible populations, to avoid favouring a narrow range of entropy values before observing the data. This is the Nemenman–Shafee–Bialek (NSB) estimator. More recently, Piga et al. (2024) estimate the strength of a Dirichlet prior from the observed counts. This strength controls how much the probability estimates are drawn towards a uniform distribution, including for unobserved categories.

Entropy estimators can also be combined to estimate MI through $I(X;Y) = H(X) + H(Y) - H(X,Y)$, as in Equation (2.5). Archer et al. (2013) study this route alongside estimators that model the joint probabilities directly. Combining three separately constructed Bayesian entropy estimates need not correspond to one joint probability model and can produce negative MI estimates. Nevertheless, these combined estimators generally outperform the authors' fully Bayesian MI estimator in their tested examples.

The sampling pattern can also suggest a more targeted estimator. Hernández and Samengo (2019) consider a well-sampled, low-entropy variable paired with a variable whose many categories are poorly sampled. They use the well-estimated marginal

and model the conditional probabilities to estimate MI, rather than estimating three entropies separately. Their results also show that combining Bayesian entropy estimates can retain bias under severe undersampling. Estimation errors need not cancel when MI estimates from two distinct populations are subtracted. The two-sample problem therefore requires both a small bias in the estimated difference and a standard error that describes its sampling variation.

2.3 Sampling variance of plug-in MI

Comparing two MI estimates requires knowing how much each varies across repeated samples. We derive the large-sample approximation

$$\mathrm{Var}\{\widehat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004).

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where a is the value around which the function is expanded and $x-a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\widehat{u} \log \widehat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u} \log \widehat{u} - u \log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \widehat{I}(P) - I(P) \approx & \sum_{i,j} (\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ & - \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ & - \sum_{j} (\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}). \end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_{j} (\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_{i} (\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_{i} \left[(\log p_{i+} + 1) \left\{ \sum_{j} (\widehat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.18}$$

The inner sum over j is the probability error for a fixed row i. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned}
&(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\
&\quad = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right) - 1 \\
&\quad = \ell_P(i,j) - 1. \qquad (2.19)
\end{aligned}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}). \qquad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j}(\widehat{p}_{ij} - p_{ij}) = \sum_{i,j}\widehat{p}_{ij} - \sum_{i,j}p_{ij} = 1 - 1 = 0. \qquad (2.21)$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}
\widehat{I}(P) - I(P) &\approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\
&= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j}(\widehat{p}_{ij} - p_{ij}) \\
&= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \qquad (2.22)
\end{aligned}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n}\sum_{a=1}^{n}\left\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\right\}. \qquad (2.23)$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\ell_P(i,j)\left[\frac{1}{n}\sum_{a=1}^{n}\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}\right]. \qquad (2.24)$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \frac{1}{n}\sum_{a=1}^{n}\left[\sum_{i,j}\ell_P(i,j)\mathbf{1}(Z_a=(i,j)) - \sum_{i,j}p_{ij}\ell_P(i,j)\right] \\ &= \frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\,. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing Z_a, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$\begin{aligned}\mathrm{E}_P\{\ell_P(X,Y) - I(P)\} &= \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\} \\ &= \underbrace{\sum_{i,j}p_{ij}\ell_P(i,j)}_{I(P)} - I(P)\underbrace{\sum_{i,j}p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The centred function $\ell_P(i,j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\}^2. \tag{2.27}$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned}\mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging n independent contributions therefore divides their variance by n. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0,1). \tag{2.29}$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

2.4 Approaches to testing MI differences

This section reviews methods for estimating uncertainty in MI and testing differences between two MI values. The question in this thesis is whether two populations have the same MI, $I(P) = I(Q)$, even when their cell probabilities differ.

The Normal Wald test.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For MI estimated from independent samples from P and Q, the first-order variance contributions add:

$$\operatorname{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term is the sampling variance of one MI estimate, and independence of the two samples makes their covariance zero. Replacing the population variances by

estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes positive population cell probabilities, positive pointwise-MI variances and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses the same standard normal distribution cutoff, regardless of how much the estimated standard error varies.

The Welch–Satterthwaite principle.

This thesis adapts the Welch–Satterthwaite principle from comparing population means to comparing population MI values. Welch's test allows the population variances to differ and compares the standardised difference between two sample means with a Student t distribution with ν degrees of freedom (Welch, 1947). For two independent samples with sample means $\bar{X}_P, \bar{X}_Q$ and sample variances s_P^2, s_Q^2, the statistic for testing equal population means and the effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both samples' degrees of freedom affect ν. Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

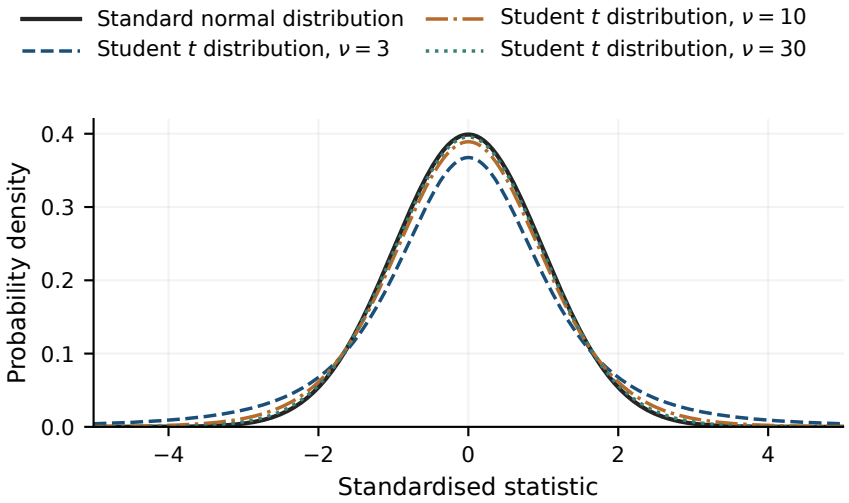


Figure 2.1: Standard normal distribution and Student t distributions with 3, 10 and 30 degrees of freedom.

Figure 2.1 shows that the Student t distribution has heavier tails than the standard normal distribution, especially when ν is small. More probability in the tails means a larger absolute statistic is needed to reject the null at the same significance level. As ν increases, the Student t distribution approaches the standard normal distribution, and their rejection cutoffs become closer.

For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. The ordinary $n_P - 1$ and $n_Q - 1$ values follow from the scaled chi-squared distribution of a sample variance under normal sampling; they do not automatically apply to an estimated MI variance. Chapter 4 therefore derives the component degrees of freedom from the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, using the same moment-matching principle.

This distinction matters because the size of a variance estimate and its stability answer different questions. The estimate $\widehat{V}(P)/n_P$ describes the expected spread of an MI estimate across samples. The variability of $\widehat{V}(P)$ describes how much that assessment of spread itself changes across samples. Two variance estimators with the same mean can have different variability. Satterthwaite's calculation uses both moments: greater variability relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite, 1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). For example,

Pan and Wall (2002) calculate approximate degrees of freedom by examining how a regression variance estimate varies between samples, and Expanded Welch uses the variability of the estimated MI variance to choose degrees of freedom in the same way. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student t distribution can widen the intervals and reduce excessive rejection in such settings.

Previous work on MI uncertainty and comparisons.

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. The first-order results of Moddemeijer (1989) and Brillinger (2004) give this uncertainty as $V(P)/n_P$, with $V(P)$ determined by the population table. Replacing its probabilities by observed proportions produces the variance estimate used by Wald. This supplies a standard error for the MI difference. Choosing a Welch–Satterthwaite cutoff requires one further quantity: how much the estimated variance varies when a new sample produces a different table.

A Bayesian approach describes uncertainty about population MI given an observed table. Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities to calculate the variance and shape of the resulting MI distribution. Our repeated-sampling comparison instead asks how often Normal Wald and Expanded Welch reject equal MI when new samples are drawn from fixed populations P and Q.

Hutcheson's test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student t distribution, using

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Hutcheson's rule assigns degrees of freedom from sample sizes. Expanded Welch instead calculates how the estimated MI variance changes with the table, including the effects of a cell change on its row and column totals.

The entropy comparison establishes that a Student t distribution can be used to compare estimated information quantities, but MI requires more than substituting a different numerator. MI combines the row, column and joint entropies of the

same observations. Their estimation errors are connected, because one cell count contributes to all three. The MI variance formulas in Section 2.3 account for that connection. Expanded Welch also tracks how the estimated MI variance changes: a cell change alters both the weights and the pointwise-MI values whose weighted variance is being estimated. Thus Hutcheson supplies a close comparison framework, while the cell and margin derivatives supply the MI-specific inputs.

In a segregation application, Mora and Ruiz-Castillo (2009) derive a normal approximation for comparing positive population MI between populations and time periods, providing a direct precedent for the Normal Wald baseline. Their later peer-reviewed account places the MI index within the segregation literature (Mora and Ruiz-Castillo, 2011).

By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based k-sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

Resampling alternatives.

A bootstrap approximates sampling variation by repeatedly drawing samples with replacement from the observed data (Efron, 1979). For this comparison, samples would be drawn separately from the two observed tables, using their original sample sizes, and the MI difference recalculated for each pair. Resampling from separate empirical probability tables allows their cell probabilities to differ and provides an estimated sampling distribution of the MI difference. The bootstrap variation around the observed estimate could then be used to construct an uncertainty interval for the population difference and assess whether that interval includes zero.

An ordinary empirical bootstrap cannot sample an unobserved cell, because it assigns that cell zero probability. It can therefore miss error arising from unobserved cells. Paninski (2003) gives a broader warning: confidence intervals based on resampling can be too narrow when information estimates remain biased.

A simple permutation test instead combines the two samples and shuffles their group labels. This gives an exact test when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Equal MI alone does not guarantee that this label-permutation test has the stated false-positive rate (Chung and Romano, 2013). Specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013); their validity for equal MI would need to be established.

2.5 Previous work and this thesis's contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson's entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose degrees of freedom. Applying this approach to MI requires calculating how the estimated MI variance changes with the observed table, including changes to the cell probabilities, row and column totals, and pointwise MI values.

This thesis derives how changes in the observed count table affect the estimated MI variance and uses that result to calculate MI-specific degrees of freedom for Expanded Welch. The resulting test retains Normal Wald's MI difference and standard error but compares the statistic with a Student t distribution rather than a standard normal distribution.

The experiment measures errors in the shared MI difference and standard error alongside false-positive rates and power. Paninski (2003) shows that finite-sample information estimates can retain substantial bias after a leading correction. Such bias or an inaccurate first-order variance approximation can undermine both tests, even when Expanded Welch lowers rejection.

Chapter 3

Overview of the Two-Sample MI Test and Wald Baseline

Normal Wald and Expanded Welch share an estimated MI difference and standard error. This chapter defines their statistic and the Normal Wald p-value.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals n_P and n_Q. The rows and columns represent the same categorical variables X and Y in both populations, with the same category definitions. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling)**.** *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets)**.** *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities)**.** *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI)**.** *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units)**.** *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Miller, 1955; Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{\widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P}\right\} - \left\{\widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q}\right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. The correction uses the declared row and column categories because every population cell has positive probability under the model, even when some cells are unobserved in a sample. Corrected estimates can be negative; those values are retained when calculating the difference. When $n_P = n_Q$, the two corrections cancel, so $\widehat{\Delta}_{\mathrm{BC}}$ is exactly the raw plug-in MI difference. With unequal sample sizes,

their difference removes the leading unequal-sample bias described in Section 2.2. A correction that estimates support from occupied cells would be a different estimator.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

The common standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of T indicates whether P or Q has the larger corrected MI estimate, while its magnitude measures the difference relative to its estimated standard error. Under H_0, we compare T with a reference distribution centred at zero. Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald test and motivation for Expanded Welch

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where Φ is the standard normal distribution function.

Why use a Student t distribution?

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the variability of that variance estimate to choose a Student t distribution instead. Hutcheson's entropy test and ordinary Welch's test set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how cell probabilities and margins make the estimated MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

Chapter 4 derives $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

Detailed Derivation of the Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3, but changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples. The name Expanded Welch refers to this expansion of the MI variance as the table probabilities change.

4.1 Choosing degrees of freedom for Expanded Welch

The Welch–Satterthwaite equation combines the two estimated MI variance contributions (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.1}$$

The terms $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$ are the estimated variance contributions used to calculate the standard error for both Normal Wald and Expanded Welch in Chapter 3. The new variables being introduced are the component degrees of freedom ν_P and ν_Q. They describe how much each estimated MI variance varies between samples relative to its mean.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\widehat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change tells us how variable $\widehat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

Moment matching for one MI variance estimate.

Satterthwaite approximates the nonnegative variance estimate $\widehat{V}(P)$ by the scaled chi-squared variable

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}. \tag{4.2}$$

The scale and degrees of freedom are chosen to match the mean and variance of $\widehat{V}(P)$.

For a chi-squared variable with ν_P degrees of freedom,

$$\mathrm{E}(\chi^2_{\nu_P}) = \nu_P, \qquad \mathrm{Var}(\chi^2_{\nu_P}) = 2\nu_P. \tag{4.3}$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.4}$$

Its variance is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P^2}\,\mathrm{Var}(\chi^2_{\nu_P}) \tag{4.5}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P}. \tag{4.6}$$

Matching this quantity to $\mathrm{Var}\{\widehat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.7}$$

The numerator depends on the mean of the estimated MI variance; the denominator measures how much that estimate fluctuates between samples. To first order, $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$, so the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$.

4.2 How the MI variance changes with the table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.8}$$

where $\delta_{(x,y)}$ places probability one on cell (x, y). At $\varepsilon = 0$, the distribution is P. Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j),

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \tag{4.9}$$

The joint and marginal probability derivatives at the original distribution are

$$\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \tag{4.10}$$

$$\left.\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \tag{4.11}$$

$$\left.\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \tag{4.12}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \tag{4.13}$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.14}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\right|_0 &= \frac{\mathbf{1}\{i=x,j=y\}-p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}-p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}-p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x,j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1. \end{aligned} \tag{4.15}$$

The joint term supplies -1, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.16}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 &= \sum_{i,j}\{\mathbf{1}(i=x,j=y)-p_{ij}\}\ell_P(i,j) \\ &\quad + \sum_{i,j} p_{ij}\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 . \end{aligned} \tag{4.17}$$

The first sum is $\ell_P(x,y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \ell_P(x,y) - I(P)}. \tag{4.18}$$

This is the centred pointwise-MI influence function from Section 2.3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y)\}. \tag{4.19}$$

Write this variance in second-moment form:

$$V(P) = \mathrm{E}_P\{\ell_P(X,Y)^2\} - I(P)^2. \tag{4.20}$$

Define the observation-level variance sensitivity by

$$g_P(x,y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.21}$$

Differentiating Equation (4.20) gives

$$g_P(x,y) = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 - 2I(P)\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.22}$$

It remains to differentiate

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.23}$$

The product rule gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 &= \sum_{i,j}\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 \ell_P(i,j)^2 \\ &\quad + 2\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.24}$$

The first sum is

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.25}$$

For the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.26}$$

$$\mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.27}$$

Substitution of Equation (4.15) gives

$$\begin{aligned}&\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 \\ &\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\ &\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P).\end{aligned} \tag{4.28}$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in

Equation (4.25) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.28) gives

$$\boxed{\begin{aligned}
g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\
+ 2\Big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\
- \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\Big].
\end{aligned}} \tag{4.29}$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for the fact that they change too: each pointwise-MI value uses a cell probability and its row and column totals. The conditional means collect the effects of those changing totals. Both parts are needed to describe how the estimated variance varies across samples.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.30}$$

The sensitivities average to zero under P, as shown in Appendix A. Their variation, not their average, determines how much the estimated MI variance changes between samples.

Assumption 6 (Regular variance sensitivity)**.** *For the first-order Expanded Welch construction,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{\delta_{Z_a^{(P)}} - P\right\}. \tag{4.31}$$

The plug-in $\widehat{V}(P)$ applies the variance calculation to this empirical table. Equation (4.21) gives its first-order change for one observation's direction. Because a first-order change is linear in the table error (van der Vaart, 1998, Chapter 20), the changes from the observations can be averaged:

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \tag{4.32}$$

The observations are independent, so

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.33}$$

$$= \frac{\tau^2(P)}{n_P}. \tag{4.34}$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \tag{4.35}$$

To interpret Equation (4.35), divide the sampling variance of $\widehat{V}(P)$ by $V(P)^2$, the square of the population pointwise-MI variance:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \tag{4.36}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When the estimate fluctuates little compared with the size of $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution, whereas greater relative fluctuation gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \widehat{g}_P(i,j) = {} & \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ & + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'} \widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j} \widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right]. \end{aligned} \tag{4.37}$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij} \left\{ \widehat{g}_P(i,j) - \sum_{k,l} \widehat{p}_{kl} \widehat{g}_P(k,l) \right\}^2 . \tag{4.38}$$

The estimated component degree of freedom is

$$\boxed{\widehat{\nu}_P = \frac{2n_P \widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.39}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Combining the two samples' degrees of freedom.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\widehat{\nu}_{\text{expanded}} = \frac{\left\{ \widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q \right\}^2}{\left\{ \widehat{V}(P)/n_P \right\}^2 / \widehat{\nu}_P + \left\{ \widehat{V}(Q)/n_Q \right\}^2 / \widehat{\nu}_Q}}. \tag{4.40}$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2\left\{1 - F_t(|T|; \widehat{\nu}_{\text{expanded}})\right\}, \tag{4.41}$$

where $F_t(x;\nu)$ denotes the cumulative distribution function of a Student t distribution with ν degrees of freedom, evaluated at x.

For a fixed finite T, the Student t distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid, so every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. Expanded Welch cannot raise rejection where Wald is already too conservative. Appendix A.5 examines a zero-component case in the combined degrees-of-freedom formula.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.42}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In N^P, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for N^Q gives $\widehat{I}(Q) = 0.25848$ nats. Since $(3-1)(3-1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.43}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.44}$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083 and -0.64436, weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

Because this count table is symmetric, the first column has the same conditional mean. Substituting these values into Equation (4.37) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P) = 0.93212$ and $\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.39) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.45}$$

Combining them with Equation (4.40) gives $\widehat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and T.
4	Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$.
5	Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, MI is zero and its first-order change is also zero. The leading changes in MI, the pointwise-MI variance $V(P)$, and the variance sensitivity $\tau^2(P)$ can therefore be compared using a second-order expansion. Appendix A.2 gives the calculation and its fixed-margin assumptions. Close to independence on that path, it gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.46}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.47}$$

This population approximation describes the estimated degrees of freedom when sampling errors are small enough for the first-order calculation to be accurate. Smaller population MI then gives fewer degrees of freedom and a larger cutoff from the Student t distribution.

Very close to independence, sampling noise itself creates apparent MI in the empirical table. The resulting variance estimates need not be close to their population values, so the estimated degrees of freedom need not tend to zero with population MI at a fixed sample size. These local expansions assume fixed table dimensions, positive cell probabilities and the fixed-margin path. Appendix A.4 retains the quadratic sampling terms needed when population MI is very small.

Theoretical status and validity conditions.

Equations (4.18) and (4.29) are exact derivatives for positive population tables and provide first-order approximations to the sampling errors in MI and its estimated variance. The scaled chi-squared model in Equation (4.2) and the Student t distribution for T are finite-sample approximations. The MI difference and standard error come from the same tables. A sample that changes the estimated difference can also change its standard error, and Expanded Welch's larger cutoff does not account for how the estimated MI difference and its standard error move together. Appendix A.3 gives the first-order covariance between each population's MI and variance estimates. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student t distribution approaches the standard normal distribution used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error, while Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. Appendix B.4 specifies the numerical criteria; the experiment evaluates validity alongside calibration.

Chapter 5

Experimental Design

The simulations draw count tables from populations with known MI. This allows us to measure errors in the estimated MI difference and its standard error, and to count false positives when the population MI values are equal. The evaluation addresses the two research questions in Section 1.2: whether the variance approximations used by Expanded Welch are accurate, and how its false-positive rates and power compare with those of Normal Wald. The design follows the simulation framework of Morris et al. (2019), specifying the populations, quantities being estimated, methods and performance measures.

5.1 Setup: Constructing joint distributions with known MI

Before drawing samples, we construct population probability tables with specified MI values. Equal MI values allow us to measure false-positive rates, while different MI values allow us to measure power. For each table, we first choose how frequently the categories occur, then change the association between X and Y while keeping those frequencies fixed.

The main simulations use square tables. Both variables have k categories, labelled $1, \ldots, k$, so the joint distribution has $k \times k$ cells. Let $a = (a_1, \ldots, a_k)$ specify the category frequencies: a_i is the probability of category i, with $a_i > 0$ and $\sum_i a_i = 1$. Within one population, we use this same vector for both variables. Thus a specifies the row margins for X and the column margins for Y.

To obtain independent variables, we could draw X using a, then draw Y independently using a. Cell (i, j) would have probability $a_i a_j$, and MI would be zero. We introduce dependence by allowing Y to copy the category drawn for X on

some observations. Let λ be the probability of copying. For each observation, first draw X using a, then choose between two ways of generating Y, independently of the value of X:

- With probability λ, set $Y = X$.
- With probability $1 - \lambda$, draw Y independently using a.

An independent draw can still match X by chance, whereas copying guarantees a match. Increasing λ therefore makes matching categories occur more often.

Both choices preserve the category frequencies of Y. When Y copies X, its categories have probabilities a because those are the probabilities used to draw X. An independent draw of Y also uses a. Consequently, changing how often we copy changes which categories occur together, while the row and column margins remain fixed. Sample row and column proportions fluctuate around these population probabilities.

To calculate a cell probability, add the contributions from the two choices. The independent choice is made with probability $1 - \lambda$ and produces $X = i, Y = j$ with probability $a_i a_j$, contributing $(1 - \lambda)a_i a_j$. The copying choice contributes only to diagonal cells, where the categories match. Reaching cell (i, i) requires both copying and $X = i$, so its contribution is λa_i. Write $\mathbf{1}_{\{i=j\}}$ for an indicator that equals one when $i = j$ and zero otherwise. Adding the two contributions gives

$$p_{ij}(\lambda) = (1 - \lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \leq \lambda < 1, \tag{5.1}$$

where the restriction $\lambda < 1$ keeps the independent choice possible and therefore gives every cell positive probability. This copying construction is a special case of the mixture model described by Bocci et al. (2010).

At $\lambda = 0$, MI is zero. Increasing λ increases MI, although the copying probability needed to reach a particular MI depends on the category frequencies a. To construct a table with the desired MI, we hold a fixed and search numerically for λ, calculating MI from the population table at each candidate value. Appendix B gives the available MI range, the numerical search and the fixed-margin calculation.

Constructing a 2×2 joint distribution with positive MI.

For population P, take $a = (0.7, 0.3)$. Each variable therefore has probability 0.7 of taking category 1 and probability 0.3 of taking category 2. Choose a target MI of 0.02 nats. The numerical search gives a copying probability of approximately 0.20347.

Cell $(1,1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, while an independent draw gives $Y = 1$ with probability 0.7. Cell $(1,2)$ also requires $X = 1$, but copying cannot give $Y = 2$. It therefore requires the independent choice followed by a draw of category 2, which has probability 0.3. Multiplying these conditional probabilities by the probability of $X = 1$ gives

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273,$$
$$p_{12} \simeq 0.7(1 - 0.20347)(0.3) = 0.16727.$$

Both off-diagonal cells have probability $(1 - \lambda)(0.7)(0.3)$, so $p_{21} = p_{12}$. The second row totals 0.3, giving $p_{22} \simeq 0.3 - 0.16727 = 0.13273$.

For population Q, use $a = (0.8, 0.2)$ and a target MI of 0.03 nats. Its numerical search gives a copying probability of approximately 0.26091. The resulting tables are

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.2}$$

These tables have $I(P) = 0.02$ and $I(Q) = 0.03$ nats, so the true signed difference is $\Delta = I(P) - I(Q) = -0.01$ nats. To construct an equal-MI comparison, we instead choose the same target MI for both populations and search for each copying probability separately.

Table sizes and margin regimes.

The experiment repeats this construction for square tables of sizes $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16, containing 4, 16, 64 and 256 cells. Each doubling of the number of categories multiplies the number of cells by four. At a fixed sample size, observations must then be distributed over more possible cells.

At each table size, we compare three patterns of category frequencies, called margin regimes. Uniform margins assign probability $1/k$ to every category. For uneven margins, the first category has probability ρ, and the remaining probability $1 - \rho$ is shared equally among the other $k - 1$ categories. We write this vector as

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \ldots, \frac{1-\rho}{k-1}\right). \tag{5.3}$$

For example, assigning probability 0.8 to the first of four categories gives

$$m_4(0.8) = \left(0.8, \frac{0.2}{3}, \frac{0.2}{3}, \frac{0.2}{3}\right).$$

Table 5.1 specifies the three regimes. Same skew uses the same uneven frequencies in both populations; different skew uses a first-category probability of 0.7 in P and

0.8 in Q. Thus the row and column margins agree within each population, while the margins can differ between populations.

Table 5.1: Row and column margins for each population.

Regime	P: rows and columns	Q: rows and columns
Uniform	$(1/k, \ldots, 1/k)$	$(1/k, \ldots, 1/k)$
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

Under equal MI, the uniform and same-skew regimes give identical joint probability tables, $P = Q$: both populations have the same margins and the same copying probability. The different-skew regime instead gives different tables with equal MI because the category frequencies differ. Comparing the regimes therefore measures false positives under both identical tables and distinct tables with equal MI.

5.2 RQ1: Checking the variance approximations

Expanded Welch chooses its degrees of freedom from the mean and sampling variance of an estimated variance. To check these quantities, we hold the population table P and sample size n_P fixed, then draw repeated count tables. Each draw gives a new $\widehat{V}(P)$, which estimates the pointwise-MI variance $V(P)$. Dividing by n_P gives that population's estimated contribution to the variance of the MI difference. The average of the $\widehat{V}(P)$ values and their variation across draws are the two moments needed for the degrees-of-freedom calculation.

The observation-level sensitivity derived in Chapter 4 describes how the estimated MI variance responds to changes in the table. Let $\tau^2(P)$ be its population variance. The first-order predictions are

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P), \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}. \tag{5.4}$$

We compare $V(P)$ with the observed mean of $\widehat{V}(P)$, and $\tau^2(P)/n_P$ with its observed sampling variance. We also average the sample-based prediction $\widehat{\tau}^2(P)/n_P$ across the repeated tables. Comparing this average with the same observed sampling variance shows the effect of estimating the sensitivity from each sample.

We next examine how the moment approximations affect the degrees of freedom. Using the actual sampling moments gives the component target

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{5.5}$$

The repeated samples estimate the mean and variance in this expression. We compare both the population first-order degrees of freedom and the average estimated component degrees of freedom with this target. For the latter, the ratio $\mathrm{mean}(\widehat{\nu}_P)/\nu_{P,\mathrm{mom}}$ equals one when the average estimate matches the target; a ratio below one means that the estimated degrees of freedom are lower on average. Medians and percentile ranges describe their sampling spread. When a summary uses only positive finite degrees of freedom, we report the included fraction. The same checks are made for Q. Appendix B.7 gives the definitions and algebra.

Table 5.2 specifies two grids. The first varies sample size at common MI 0.02 nats. The second includes very small positive MI to examine behaviour near independence, where the first-order variance becomes small and quadratic sampling terms can matter. Both grids use all three margin regimes and equal sample sizes $n_P = n_Q = n$. Each row combines every listed table size, MI value and sample size.

Table 5.2: Settings for the approximation checks and broad null comparison.

Check	Parameter values
Across sample sizes	$k \in \{2, 4, 8, 16\}$, $I(P) = I(Q) = 0.02$ nats. $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Near independence	$k = 4$, $n \in \{100, 1000, 10000\}$. $I(P) = I(Q)$ takes the values (nats) $0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05$.

5.3 RQ2: False-positive rates under equal MI

To measure false positives, we set the two population MI values equal and repeatedly test sampled table pairs. Every rejection is then a false positive. Normal Wald and Expanded Welch use the same bias-corrected MI difference and estimated standard error from Chapter 3. Their cutoffs differ: Normal Wald uses the standard normal distribution, while Expanded Welch uses a Student t distribution with MI-specific degrees of freedom estimated from that pair.

Both tests use a two-sided 5% nominal test. The broad null comparison uses the first grid in Table 5.2: common MI 0.02 nats, four table sizes, three margin regimes

and ten sample sizes, giving 120 settings. At each setting, a false-positive rate above 5% means excessive rejection; a rate below 5% means conservative rejection. We also report the fraction of valid results, because an unusable calculation contributes no rejection.

The fixed moment reference checks whether a Student t distribution based on the denominator's sampling moments gives reliable null rejection. It uses the same MI difference and standard error as both tests. Its degrees-of-freedom target uses the mean and sampling variance of the *combined* squared standard error, $\widehat{\mathrm{SE}}^2$:

$$\nu_{\mathrm{ref}} = \frac{2\{\mathrm{E}(\widehat{\mathrm{SE}}^2)\}^2}{\mathrm{Var}(\widehat{\mathrm{SE}}^2)}. \tag{5.6}$$

Independent null simulations estimate these moments, and the fitted degrees of freedom stay fixed throughout evaluation at that setting. Expanded Welch instead recalculates its degrees of freedom for every sampled pair. Matching the denominator moments does not by itself ensure that the shared statistic follows a Student t distribution. Appendix B.7 gives the moment calculation.

5.4 RQ2: Nominal and calibrated power

Power is the probability of rejecting equal MI when the population MI values differ. We first compare power using the nominal 5% cutoffs. A test that rejects less often under equal MI can also reject less often under alternatives, so we then compare power after adjusting the cutoffs to match rejection at a specified equal-MI population.

The focused comparison uses settings A–C in Table 5.3. Earlier simulations identified A and B as settings where Expanded Welch brought false-positive rates closer to 5%; their power is evaluated on new independent samples. Setting C provides a larger-sample comparison.

To choose the MI differences for a power curve, we first measure the sampling variation under equal MI. An independent null simulation estimates the standard deviation (SD) of the corrected MI difference; denote this fitted value by σ_0. This fitting batch also fixes ν_{ref} for the entire curve. Measuring differences in units of σ_0 makes the three curves comparable relative to their null sampling variation.

Table 5.3: Focus settings for the power comparison; common null MI is 0.02 nats.

Setting	Margins	$n_P = n_Q$	Size	Fitted null SD σ_0 (nats)
A	Uniform	20	4×4	0.154967
B	Different skew	100	8×8	0.065138
C	Different skew	10,000	4×4	0.003175

For each curve, we hold $I(P) = 0.02$ nats and increase $I(Q)$. Let h be the number of fitted null SDs separating the population MI values. The populations satisfy

$$\begin{aligned} I(P) = 0.02, \qquad I(Q) = 0.02 + h\sigma_0, \\ h \in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}, \\ \Delta = I(P) - I(Q) = -h\sigma_0. \end{aligned} \tag{5.7}$$

Thus $h = 0$ gives equal MI, while positive h gives larger MI in Q and a negative signed difference. For example, at C, two null SDs are approximately $2(0.003175) = 0.00635$ nats. The corresponding alternative has $I(P) = 0.02$, $I(Q) \simeq 0.02635$ and $\Delta \simeq -0.00635$ nats. The fitted SD and the resulting population tables stay fixed during evaluation. Appendix B gives their construction and the fixed SD values.

Power after matching null rejection.

Empirical calibration chooses a p-value threshold separately for each test and the fixed moment reference. For each full power curve, the procedure is:

1. Use the independent fitting samples with $I(P) = I(Q) = 0.02$ nats, retaining the curve's margins, table size and sample sizes.
2. Choose each rule's threshold to give at most 5% rejection in that fitting batch, and hold it fixed across the curve.
3. Apply the thresholds to separate equal-MI evaluation samples and to the alternative samples, measuring the achieved false-positive rates and power.

Normal Wald and the fixed moment reference give identical calibrated decisions because each orders samples by $|T|$: with fixed degrees of freedom, larger $|T|$ always gives a smaller p-value. Expanded Welch can order samples differently because its degrees of freedom change with the sampled pair.

Changing the calibration population.

Equal MI can occur at different common MI values. To examine how that choice affects the power comparison, we repeat calibration at $h = 2$ and $h = 4$ for A–C using the two null populations in Table 5.4.

The lower null sets both MI values to the alternative's smaller value; the upper null sets both to its larger value. Both retain the alternative's margins and sample sizes.

Table 5.4: The alternative and two calibration nulls; MI values are in nats.

Population pair	$I(P)$	$I(Q)$
Fixed alternative	0.02	$0.02 + h\sigma_0$
Lower null	0.02	0.02
Upper null	$0.02 + h\sigma_0$	$0.02 + h\sigma_0$

Independent upper-null batches fit and validate the new cutoffs. We apply both calibrations to the same alternative pairs, so any change in measured power comes from the cutoffs rather than a change of alternative samples. Each calibration is also evaluated at both nulls. Matching rejection at one null population does not ensure the same false-positive rate elsewhere under equal MI. Appendix B.10.1 gives the construction and uncertainty calculation.

Three supporting comparisons extend the focused power curves:

- **Broader MI differences:** differences from 0 to 0.2 nats at $n = 1{,}000$ and 10,000.
- **Association construction:** replace copying by a probability transfer within four cells, preserving the margins and target MI. This changes where dependence occurs in the table.
- **Sample allocation:** keep each population pair and 2,000 observations fixed, comparing allocations (n_P, n_Q) of $(1000, 1000)$, $(400, 1600)$ and $(1600, 400)$.

Appendix B gives the exact settings; the experimental supplement retains the full results.

5.5 Shared evaluation: Sampling and uncertainty

At one evaluation setting, each repetition draws a count table from P and an independent count table from Q. Both tests and the fixed moment reference use this same pair, so their rejection decisions can be compared on the same sampled data. Each batch contains 100,000 independent pairs. The study uses three types of batch:

1. **Base evaluation:** 366 distinct settings for estimating rejection rates and the performance measures below.

2. **Independent fitting:** 186 distinct equal-MI settings for moment references and calibration cutoffs.

3. **Calibration sensitivity:** six upper-null settings, each with separate fitting and validation batches.

Separate random streams supply fitting and evaluation samples. Settings shared between comparisons are sampled once within each role. The base study uses 55.2 million pairs; sensitivity adds 1.2 million.

Table 5.5 defines the measures for each evaluation setting.

Table 5.5: Performance measures calculated from repeated samples.

Measure	Calculation
Rejection rate	$\frac{\text{number of valid rejections}}{\text{number of attempted pairs}}$
Valid-result fraction	$\frac{\text{number of valid outputs}}{\text{number of attempted pairs}}$
Difference bias	$\text{mean}(\widehat{\Delta}_{\text{BC}}) - \Delta$
Variance ratio	$\frac{\text{mean}(\widehat{\text{SE}}^2)}{\text{observed variance of } \widehat{\Delta}_{\text{BC}}}$
Direction under alternatives	Correct when $\widehat{\Delta}_{\text{BC}}\Delta > 0$; opposite when $\widehat{\Delta}_{\text{BC}}\Delta < 0$.

Bias uses finite MI differences even when a test is invalid. The variance ratio uses matched finite pairs. A ratio of one indicates agreement. Ratios below one mean that the squared standard error understates the observed difference variance on average; ratios above one indicate overstatement. Power includes opposite-direction rejections, so we report estimated direction separately. Appendix B.5 specifies the inclusion rules and complete calculations.

Rejection rates have Wilson 95% confidence intervals (Wilson, 1927), which describe uncertainty from the finite number of simulation pairs. Intervals for method differences account for the shared pairs. For calibrated power, we resample fitting and evaluation outputs and refit the cutoffs, so the intervals also include cutoff-fitting uncertainty. We compare power-difference intervals with a practical tolerance of ± 1 percentage point. These are pointwise intervals for one rate or comparison at a specified setting. Appendix B gives the interval calculations.

Chapter 6

Results

Each evaluation setting uses 100,000 sampled pairs. The nominal rates divide valid rejections by all attempts, and the plots show validity separately. Brackets around rates and shading in rejection plots give pointwise 95% intervals. Intervals for the fixed moment reference condition on its independently fitted degrees of freedom.

6.1 Accuracy of the variance approximations

6.1.1 Mean and sampling variance

The population first-order approximations can predict the sampling variance of $\widehat{V}$ reasonably accurately while underestimating its sampling mean. At $n = 1{,}000$, the sampling-variance prediction divided by the observed variance ranges from 0.710 to 0.996 across all table sizes and margin regimes. Over the same grid, the observed mean divided by V ranges from 1.023 to 4.571. The mean discrepancy is particularly large for the larger tables.

Figure 6.1 shows the different-skew populations at common MI 0.02 nats. The top row compares the mean of $\widehat{V}$ with V. In the middle and bottom rows, solid curves use the population first-order predictions. Dashed curves use the mean plug-in sampling-variance prediction in the middle row and the mean estimated component degrees of freedom in the bottom row. The latter are shown only when all component estimates are positive and finite. A ratio of one indicates agreement; shading gives approximate 95% Monte Carlo intervals around the solid curves.

Increasing sample size improves both approximations. At $n = 50{,}000$, the sampling-variance ratios across the same broad grid lie between 0.974 and 1.009, while mean ratios range from 1.000 to 1.091. The experimental supplement provides the companion uniform and same-skew plots.

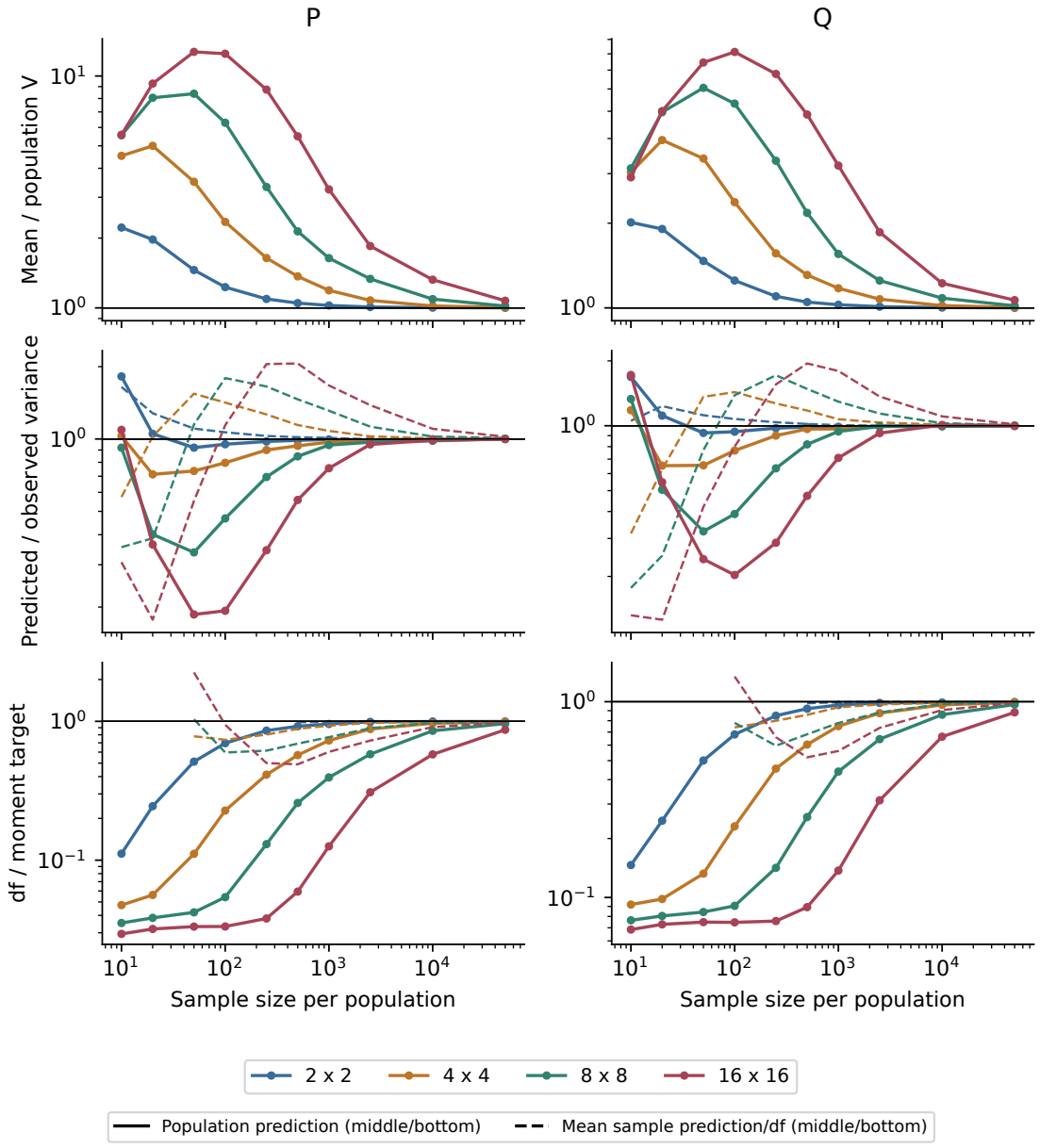


Figure 6.1: Component approximation ratios under different-skew margins.

The remaining mean error lowers the population first-order degrees of freedom relative to the sampling-moment target. For different-skew 4×4 populations at $n = 1{,}000$ and MI 0.02, the sampling-variance ratios are 0.970 for P and 0.967 for Q. The observed means are $1.190V(P)$ and $1.174V(Q)$. The first-order calculation substitutes $V(P)$ for the larger sampling mean $\mathrm{E}\,\widehat{V}(P)$, and likewise for Q. Because the moment-matching formula squares the mean, this substitution lowers the degrees of freedom even though the sampling-variance prediction is close. The resulting first-order values, 15.30 and 13.61, fall below the moment targets, 21.03 and 18.14. Table 6.1 gives the component values; Appendix B.7 gives the algebra.

Table 6.1: Component moment and degrees-of-freedom comparisons at 4×4, $n = 1{,}000$, different skew.

MI, pop.	Mean ratio	Pop. var.	Plug-in var.	Mean df	Moment df	FO df
0.0001, P	44.939	0.047	1.965	4.86	9.12	0.10
0.0001, Q	41.996	0.052	2.142	4.44	8.63	0.09
0.02, P	1.190	0.970	1.080	19.55	21.03	15.30
0.02, Q	1.174	0.967	1.072	17.00	18.14	13.61
0.05, P	1.066	1.008	1.013	42.56	43.07	37.63
0.05, Q	1.057	1.031	1.029	37.59	38.61	33.50

In the table, mean ratio is $\mathrm{E}\,\widehat{V}/V$; the next two columns divide the population and mean plug-in sampling-variance predictions by the observed sampling variance of $\widehat{V}$. Mean df averages the positive finite component estimates; all 100,000 estimates are included in each shown row. Moment df uses the actual sampling moments, while FO df uses the population first-order formula.

Estimating the component degrees of freedom from each sample can produce a further shortfall from the moment target. At setting A (uniform $4{\times}4$, $n_P = n_Q = 20$), the mean for the P component is about 6.52 against a moment target of 14.81, or 44% of the target. In the 16×16, $n = 1{,}000$ same- and different-skew cases, the corresponding ratios range from 55% to 60%. All 100,000 component estimates are included in these comparisons. At C (different-skew 4×4, $n_P = n_Q = 10{,}000$), the ratios are about 99%. The bottom-row dashed curves show this additional discrepancy when population probabilities in the first-order formula are replaced by sample proportions. The moment target itself does not guarantee an exact Student t distribution for the test statistic.

6.1.2 Behaviour near independence

Near independence, quadratic sampling terms contribute substantially to $\widehat{V}$. For 4×4 different-skew populations at common MI 0.0001 and $n = 1{,}000$, the local calculation in Appendix A.4 predicts $n\,\mathrm{E}\,\widehat{V} = 9.2$ for each population; the observed value for P is 9.158. The mean estimated component degrees of freedom for P are 4.86, close to the local prediction 4.6, while the sampling-moment target is about nine. Thus the local calculation describes the mean estimated degrees of freedom accurately even though that mean remains below the moment target.

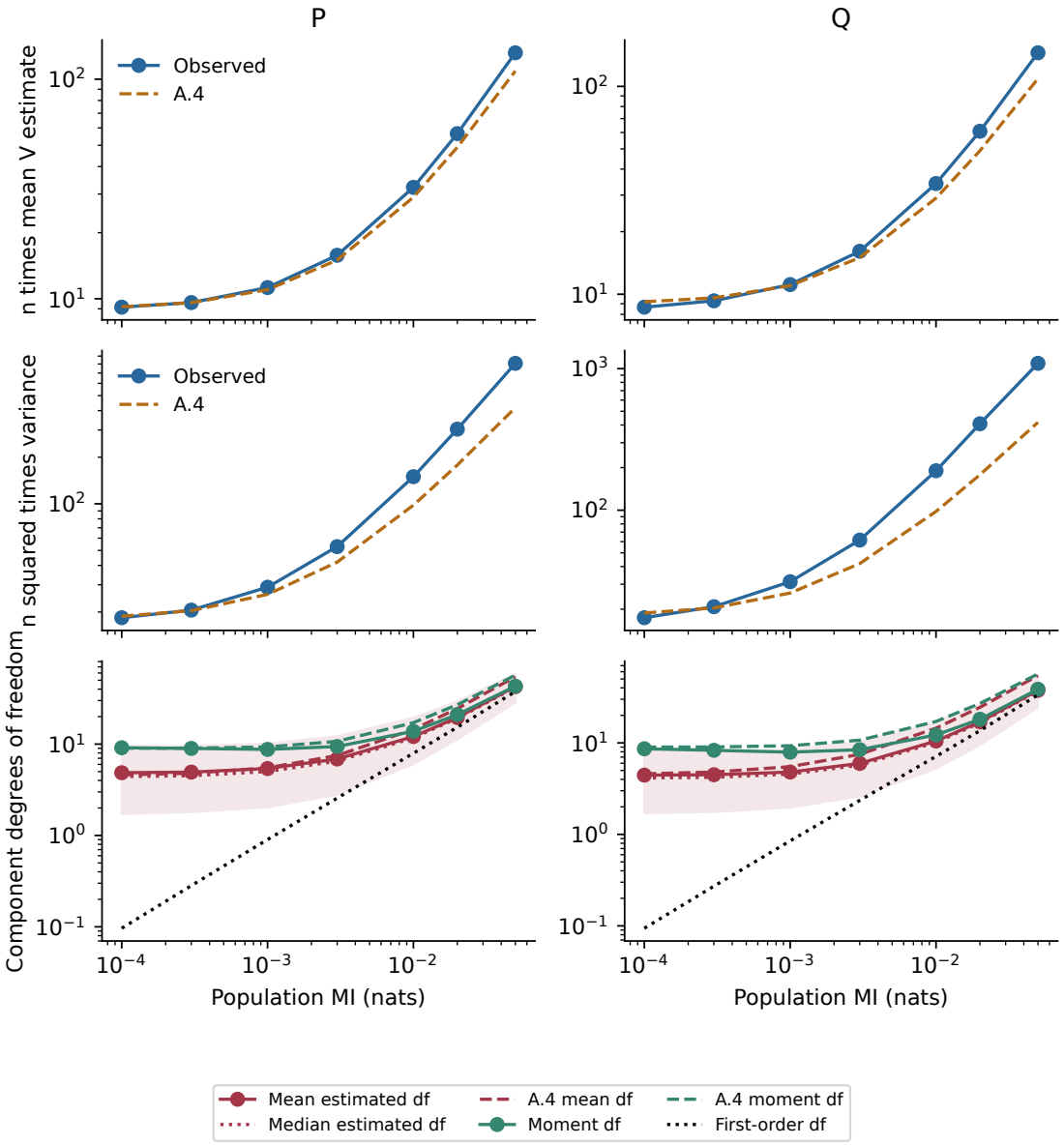


Figure 6.2: Observed component moments and Appendix A.4 predictions across MI.

The population and sample-based first-order sampling-variance predictions have errors in opposite directions near independence. At the smallest MI target in

Figure 6.2, the population sampling-variance prediction is only about 5% of the observed value. The mean plug-in prediction is 1.97 times the observed value for P and 2.14 times for Q. The quadratic sampling contribution remains substantial, while the population first-order degrees of freedom are below 0.1. The shaded band shows the fifth–95th percentile range of estimated component degrees of freedom, with their mean and median shown separately.

At MI 0.02, the population sampling-variance approximation is much more accurate, while the mean plug-in variance predictions remain about 7–8% above the observed values. The remaining mean error still lowers the first-order degrees of freedom relative to the moment target. Appendix C.1 reports the mean error across MI and margins.

6.2 False positives when MI is equal

Expanded Welch improves false-positive rates in selected settings where Normal Wald rejects too often. In others, it further lowers already conservative rates. Across the 120 settings at common MI 0.02 nats, Expanded Welch moves rejection closer to 5% in 29 settings and farther away in 52, using changes in distance greater than 0.3 percentage points. All 29 improvements occur with Wald above 5%; 48 of the 52 deteriorations occur with Wald below 5%. The remaining 39 settings fall within this tolerance.

Figures 6.3 and 6.4 show uniform and different-skew margins, with the fixed moment reference as a diagnostic. The lower panels give the fraction of valid outputs for Normal Wald and Expanded Welch. The fixed moment reference shares Wald's validity in these settings. Validity is lowest in the smallest samples: across the 120 settings, the minimum valid fractions are 0.958 for Normal Wald and the fixed reference, and 0.631 for Expanded Welch. The supplement reports conditional rates and included counts separately.

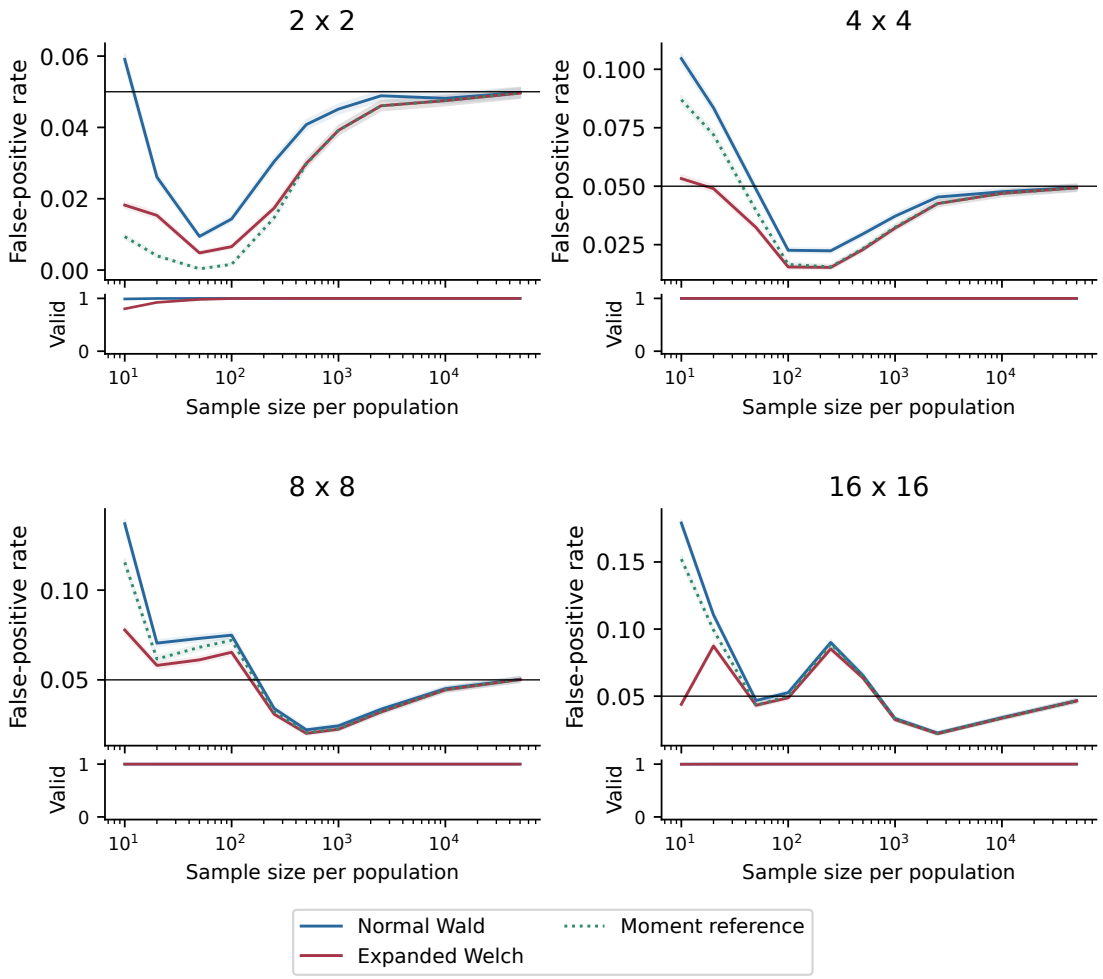


Figure 6.3: False-positive rates under uniform margins.

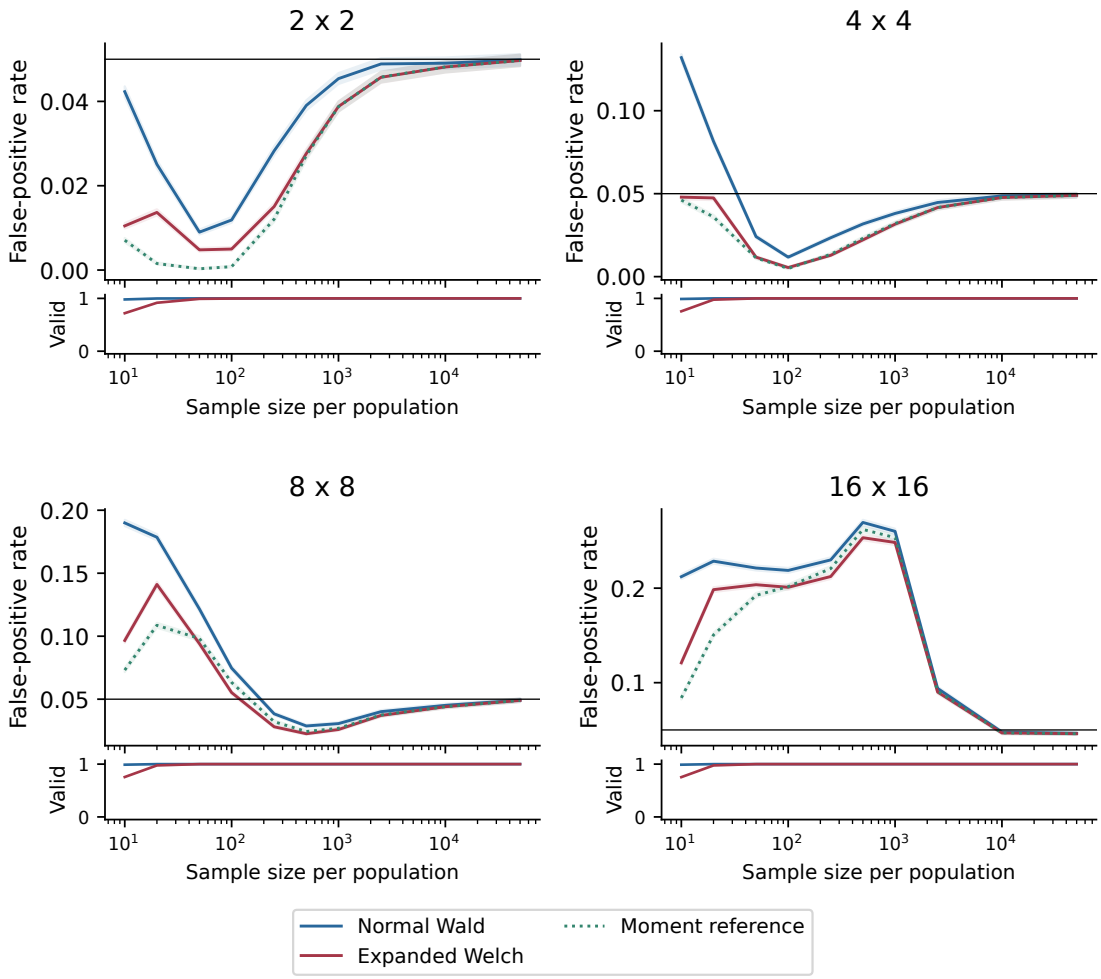


Figure 6.4: False-positive rates under different-skew margins.

6.2.1 Small-sample improvements and sparse-table failures

The selected small-sample improvements occur with different estimation errors. At the uniform 4×4 setting A, Expanded Welch gives 4.90% rejection compared with 8.35% for Wald. At the 8×8 different-skew setting B, the rates are 5.52% and 7.47%, respectively. Expanded Welch is closer to 5% at B but still rejects too often. Table 6.2 compares these settings with the larger-sample and sparse-table cases. Both tests and the fixed moment reference are valid on every sampled pair in all seven settings.

Both tests over-reject at stronger common MI: in A at 0.33 nats, Wald rejects 16.16% and Expanded Welch rejects 13.56% of pairs. In B at 0.28 nats, the rates are 7.62% and 7.25%, respectively.

Table 6.2: Null rejection rates with pointwise 95% intervals.

Setting	Normal Wald	Expanded Welch	Moment reference
A	8.35% [8.18, 8.53]	4.90% [4.76, 5.03]	7.20% [7.04, 7.36]
B	7.47% [7.30, 7.63]	5.52% [5.38, 5.67]	6.33% [6.18, 6.48]
C	4.87% [4.73, 5.00]	4.77% [4.64, 4.90]	4.77% [4.64, 4.90]
16, same, 1,000	0.60% [0.56, 0.65]	0.53% [0.48, 0.57]	0.56% [0.52, 0.61]
16, same, 10,000	4.38% [4.26, 4.51]	4.26% [4.14, 4.39]	4.27% [4.15, 4.40]
16, different, 1,000	26.01% [25.74, 26.28]	24.84% [24.57, 25.10]	25.35% [25.08, 25.62]
16, different, 10,000	4.78% [4.65, 4.91]	4.68% [4.55, 4.82]	4.70% [4.57, 4.83]

At A, difference bias is negligible, but the mean squared standard error is only 0.827 of the observed difference variance, so the standard error understates sampling variation. At B, difference bias is positive and the variance is overestimated. Expanded Welch lowers false positives in both cases, although the shared estimates have different errors. At the larger-sample setting C, bias is negligible and the variance ratio is 1.010. Table C.2 gives the numerical diagnostics and degrees-of-freedom comparisons in the appendix.

Changing the degrees of freedom does not resolve the sparse-table failure. At 16×16, different skew and $n = 1{,}000$, the difference bias is about 0.030 nats, or 2.06 observed SDs. Rejection ranges from 24.8% to 26.0% across both tests and the fixed moment reference; the reference rate is 25.3% despite its independently fitted moments. Under the equal-MI null, the estimated difference is centred at

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx 0 + 0.02985 = 0.02985 \text{ nats.} \tag{6.1}$$

This shift is large relative to the observed SD of approximately 0.01450 nats. It moves most rejections into the positive tail even though the population difference is zero.

Under equal MI, the same-skew populations have identical tables. With equal sample sizes, their expected MI estimation biases therefore cancel in the difference. Difference bias is negligible, but the variance ratio is 1.93; the three rejection rates lie between 0.53% and 0.60%. The matched populations distinguish rejection driven by a large difference bias from conservative behaviour when that bias cancels. At $n = 10{,}000$, both skew regimes have much smaller estimation errors and rates closer to 5%. The fixed moment reference's different rates across these populations show that matching denominator moments alone does not ensure a correctly calibrated Student t reference for the shared statistic.

6.3 Power at the selected settings

6.3.1 Nominal power

Expanded Welch has lower nominal power than Normal Wald at the selected small-sample settings A and B (Table 5.3). At A, two null SDs correspond to about 0.310 nats: Expanded Welch rejects 35.9% of pairs, compared with 41.0% for Wald. Figure 6.5 shows all three focus settings with MI differences measured in fitted null standard deviations. The nominal comparison includes their different null rejection rates at zero. Appendix C.2 gives numerical rates and paired contrasts.

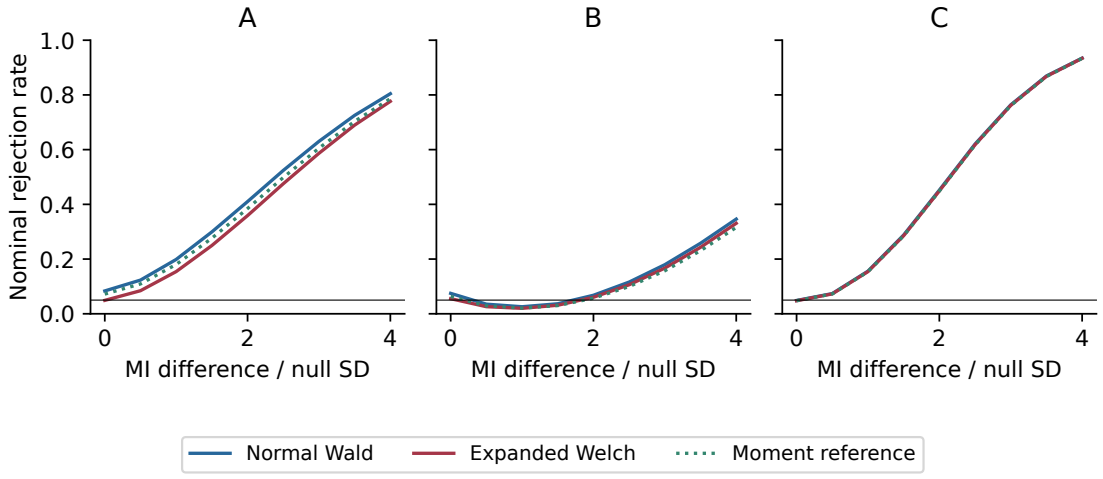


Figure 6.5: Nominal power across differences measured in fitted null standard deviations.

As the population MI values move farther apart at B, rejection initially falls before rising. Expanded Welch's rate falls from 5.52% under the null to 1.99%

at one null SD, before rising to 33.1% at four SDs. As $I(Q)$ increases, the true signed difference becomes negative, initially offsetting the positive estimation bias observed under the null. The estimated difference moves towards zero, reducing $|T|$ and rejection before the curve rises. This distortion of the shared statistic remains when the rejection cutoff is increased. Appendix C reports the estimated directions among rejections in the sparse-table comparisons.

At C, nominal power differences are small. At two null SDs, or about 0.00635 nats, Wald gives 45.2% and Expanded Welch gives 44.9%. Expanded Welch's paired loss relative to Wald is 0.30 percentage points, with a 95% interval of 0.27–0.34. Table C.4 reports paired comparisons at two null SDs for all three focus settings. The intervals use the same sampled pairs to estimate the difference directly. The calibrated comparison then assesses power after matching null rejection.

6.3.2 Power after empirical null calibration

Expanded Welch's power gain depends on which equal-MI population supplies the cutoff. With both population MI values set to 0.02 nats, the lower-null calibration gives rejection rates of 4.93% to 4.99% on independent null samples across A–C for both tests and the fixed reference. Wald and the fixed moment reference have identical calibrated decisions. Appendix Figure C.1 shows the full curves under this lower-null calibration.

The upper null sets both MI values to the fixed alternative's larger MI. Table 6.3 compares lower- and upper-null calibration while holding each alternative fixed. At A and two null SDs, the gain over Wald changes from 3.58 percentage points to a loss of 0.76 points. At B and four SDs, a 3.92-point gain becomes a 0.39-point loss. Under upper-null calibration, A at four SDs retains a small positive estimate whose interval includes zero, while B at two SDs retains a much smaller gain. At C, both calibrations give differences well within the one-percentage-point tolerance.

The second column gives the common MI of the upper null. Brackets give paired 95% bootstrap intervals including uncertainty from fitting the cutoffs.

Appendix C.2 reports the cross-null rates and full power tables. The two checks do not establish control over every population satisfying equal MI.

B's initial fall in rejection also remains after lower-null calibration for all the compared rules. Calibration leaves the shared biased difference unchanged, so the fitted power gain coexists with the distortion of the power curve.

Table 6.3: Expanded Welch minus Wald power under two null calibrations.

Setting: SD	Upper MI	Lower-null gain (pp)	Upper-null gain (pp)
A: 2	0.3299	+3.58 [+3.32, +3.80]	-0.76 [-0.91, -0.56]
A: 4	0.6399	+4.01 [+3.79, +4.20]	+0.18 [-0.12, +0.43]
B: 2	0.1503	+1.11 [+1.03, +1.19]	+0.05 [+0.01, +0.10]
B: 4	0.2806	+3.92 [+3.77, +4.14]	-0.39 [-0.48, -0.28]
C: 2	0.0263	+0.04 [+0.01, +0.07]	-0.06 [-0.06, -0.01]
C: 4	0.0327	+0.02 [+0.01, +0.03]	-0.01 [-0.03, -0.01]

6.4 Association pattern and sample allocation

Concentrating dependence in four cells reproduces the main sparse-table failure. At 16×16, $n = 1{,}000$ and different skew, the null difference bias is 0.03155 nats, compared with 0.02985 under copying. Both tests and the fixed moment reference reject far too often. With same-skew populations, expected difference bias cancels and rejection remains conservative.

In the different-skew 4×4 comparison, allocating fewer observations to Q lowers Expanded Welch's power. Assigning only 400 of the 2,000 observations to Q increases the observed SD from 0.01173 to 0.01611 nats. At the 0.01-nat alternative, power falls from 10.26% under equal allocation to 3.42%. Expanded Welch's median degrees of freedom also fall, increasing the rejection cutoff. Reversing which population has fewer observations changes the result because the two populations have different variances and residual biases. The experimental supplement reports the complete construction and allocation comparisons.

6.5 Application to online shopping data

The Online Shoppers Purchasing Intention dataset provides a practical comparison of month–purchase MI between new and returning visitors (Sakar and Kastro, 2018). A larger MI means that the month provides more information about whether a session ends in a purchase. A second comparison uses the weekend indicator instead of month to examine whether its MI with purchasing differs between the same visitor groups. This application was selected from an initial exploration of six comparisons across three public datasets.

The dataset contains 12,330 sessions, with each session belonging to a different user according to the source documentation (Sakar and Kastro, 2018). The analysis retains all 1,694 sessions labelled `New_Visitor` and 10,551 labelled

`Returning_Visitor`, excluding the 85 `Other` sessions. The outcome is the original Boolean `Revenue` variable: true indicates a purchase and false indicates no purchase. The tests treat the two disjoint visitor groups as independent samples and the sessions within each group as independent observations.

Each group forms a 10×2 month–purchase count table. The rows are the ten months recorded in the dataset: February, March, May, June, July, August, September, October, November and December. The columns are no purchase and purchase, and each session adds one to its corresponding cell. Both groups use these same categories, with no smoothing or removal of zero cells. All ten months occur in both groups, although the new-visitor February row contains only one session and no purchase. Replacing month with the Boolean `Weekend` variable gives a 2×2 table for each group, with weekday and weekend rows; all four cells are observed.

Both methods use the bias-corrected MI estimates in Table 6.4. For the month tables, the leading bias correction subtracts $(10-1)(2-1)/(2n) = 9/(2n)$ from each plug-in MI estimate (Miller, 1955; Moddemeijer, 1989); for the weekend tables, it subtracts $1/(2n)$. Here n is the number of sessions in the relevant visitor group. The estimated difference is new-visitor MI minus returning-visitor MI for the same predictor–purchase pair. Both tests use this difference and the shared standard error, with two-sided rejection at the 5% level.

Table 6.4: MI estimates and two-sample test results for online shopping sessions.

	Corrected MI (nats)			p-value	
Predictor	New	Returning	Expanded Welch degrees of freedom	Normal Wald	Expanded Welch
Month	0.004316	0.017880	26.79	4.71×10^{-5}	3.73×10^{-4}
Weekend	0.000664	0.000727	2.18	0.9554	0.9601

Month has higher estimated MI with purchasing among returning visitors. The corrected difference is -0.013563 nats, with a standard error of 0.003333 nats, giving

$$T \approx \frac{-0.013563}{0.003333} \approx -4.069. \tag{6.2}$$

Both tests reject equal month–purchase MI for the visitor groups, although the Student t distribution with 26.79 degrees of freedom gives a larger p-value than the standard normal distribution. Holm's adjustment for all six comparisons (Holm, 1979) gives month p-values of 0.000189 for Normal Wald and 0.00149 for

Expanded Welch, so rejection persists. Table C.7 gives all six comparisons and their raw and adjusted p-values.

For weekend, both corrected MI estimates are approximately 0.0007 nats. Their difference, -0.000063 nats, is small relative to the standard error of 0.001124 nats, giving $T \approx -0.056$. Both p-values remain close to one despite Expanded Welch's 2.18 estimated degrees of freedom. Neither test provides evidence of different weekend–purchase MI between the visitor groups; this does not establish equality. Small estimated degrees of freedom with weak associations and large samples are consistent with the near-independence simulations. The unknown population tables prevent a direct check of the sampling-moment target.

Chapter 7

Implications and Limitations

7.1 Accuracy of the variance approximation

The experiment shows why the two moment approximations need separate checks. The predicted sampling variance of $\widehat{V}$ can be close to its observed value while the mean estimate remains above the population variance coefficient. In the 4×4 different-skew example at $n = 1{,}000$, the variance predictions are within about 3–4%, but the mean estimates are about 17–19% too large. The moment-matching formula squares the mean. Replacing that inflated sampling mean by the smaller population value therefore lowers the population first-order degrees of freedom, even when the sampling-variance approximation is reasonably accurate. Mean estimated component degrees of freedom are lower still at A, about 44% of the moment target, compared with about 99% at C.

Near independence, the population first-order variance is small and quadratic sampling terms remain substantial. The calculation in Appendix A.4 accounts for their contribution to the observed component means and degrees of freedom. This explains why increasing sample size and moving away from independence can change which approximation is responsible for the error. It also shows why accurate population formulas do not by themselves ensure that substituting sample probabilities gives accurate degrees of freedom.

A second-order mean correction could address part of this discrepancy. The observed variation across MI and margins means that its coefficient should be derived for the population table. A fixed coefficient based only on table size would leave errors in some settings, and any estimated correction would require evaluation of the resulting test.

7.2 False-positive rates and power

Expanded Welch brings false-positive rates closer to 5% in the two selected small-sample populations, both at common MI 0.02 nats. Every rule returns a valid result there, so the lower rejection rates reflect the cutoffs rather than failed calculations. Across the broad grid, Expanded Welch also reduces rates that are already below 5%. The larger cutoff is useful when it corrects excessive rejection and costly when it deepens conservatism.

The useful null behaviour does not establish that the estimated degrees of freedom accurately match denominator moments. At A, the standard error understates the observed variation of the MI difference, while the low estimated degrees of freedom increase the rejection cutoff. These effects can compensate in the null rejection rate. At B, the variance ratio exceeds one and the difference is biased, so the same explanation cannot be carried over unchanged. Separating these contributions would require changing the individual estimates while holding the others fixed.

Calibrated power also depends on the equal-MI population used to fit the cutoff. At common MI 0.02, Expanded Welch gains power over Wald at A and B. Moving calibration to the upper equal-MI population while holding the alternative fixed reverses the gain at A and two null SDs, and at B and four SDs. Cross-null evaluation shows why matching the lower null is insufficient: the calibrated rules can still have different false-positive rates at the upper null. The sample-dependent degrees of freedom change the ordering of samples, but the observed power gain is conditional on the calibration population. At C, the differences remain practically small under both calibrations.

B retains its initial decline in rejection as the true MI difference increases. Its positive null bias initially offsets the negative change in the true difference, moving the estimated difference towards zero. Adjusting the cutoff leaves that distortion in all the compared rules. Neither a calibrated gain nor a lower nominal rejection rate resolves the bias in the shared statistic.

7.3 Estimation errors that a cutoff cannot correct

The sparse 16×16 different-skew comparison remains unreliable. Its positive difference bias is larger than the true negative difference at the 0.01-nat alternative, so almost every rejection gives the wrong estimated ordering. The calculation in Equation (C.1) explains the sign reversal. The four-cell construction reproduces the failure, and the fixed moment reference also rejects too often. Fitting denominator moments cannot remove bias from the numerator.

The same-skew null illustrates a separate error. Identical population tables at equal sample sizes cancel the expected difference bias, but the estimated difference variance is almost twice its observed value in the sparse copying case. Both tests are conservative. Unequal sample sizes can also leave unequal residual biases in otherwise identical populations. The allocation check shows how changed bias, variance contributions and degrees of freedom affect the result when one population receives fewer observations.

For an observed application, the MI difference gives estimated size and direction, while the standard error describes estimated sampling variation. Report these with the p-value and, for Expanded Welch, the estimated degrees of freedom. The online-shopping example illustrates that interpretation, but its unknown population MI values prevent a direct check of false-positive rates. Simulations from plausible application populations can examine sensitivity to rare cells, unequal sample sizes and distinct equal-MI tables. Their conclusions depend on the populations chosen for that assessment.

7.4 Limits of the evidence

The power settings were selected using earlier false-positive results and evaluated on independent samples. Each curve keeps $I(P) = 0.02$ and increases $I(Q)$ along the copying family; other directions and association patterns remain untested. The two calibration checks do not bound rejection across the entire equal-MI null. The allocation check is restricted to the specified 4×4 populations and three allocations.

Both constructions use equal row and column margins within each table and align association locations between populations. Unequal margins within a table and other locations of dependence remain untested. The pointwise confidence intervals quantify simulation uncertainty for the specified populations.

The fixed reference and empirical calibration require known simulation populations. Replacing sample-dependent degrees of freedom by a fixed value changes both their magnitude and their dependence on the statistic, so that comparison does not isolate estimation error. The calibrated power intervals include fitting and evaluation uncertainty for the fixed constructed alternatives.

The derivation assumes independent categorical observations, fixed categories, positive population cell probabilities and positive first-order MI variance. Paired samples would require a covariance term in the difference variance. Growing alphabets and continuous estimators have different approximation errors. Exact independence needs the second-order argument in Appendix A.4.

Chapter 8

Conclusion

8.1 Contributions and main findings

This thesis derives an Expanded Welch–Satterthwaite test for comparing categorical MI across independent populations. Its mathematical contribution is the observation-level sensitivity of the estimated MI variance, accounting for changes in cell probabilities, margins and pointwise MI. The variance of this sensitivity gives a first-order sampling-variance approximation, which is used with moment matching to choose MI-specific degrees of freedom for a Student t distribution.

The first research question concerns the approximations used to choose those degrees of freedom. The sampling-variance prediction can be accurate while the mean estimate remains inflated. Replacing that mean by the population variance coefficient then lowers the first-order degrees of freedom relative to the target calculated from the sampling moments. The mean estimated component degrees of freedom can fall further below that target, with the size of the discrepancy depending on the population and sample size. Near independence, the quadratic sampling contribution explains the scale of the observed component estimates and their departure from the first-order values.

For the second research question, Expanded Welch improves false-positive rates relative to Normal Wald in selected small-sample populations at common MI 0.02 nats, with a nominal power cost. Its calibrated power advantage is sensitive to the fitting population: moving calibration to the upper equal-MI population reverses the advantage in some fixed-alternative comparisons. These gains therefore describe power at a specified calibration population rather than a stable ranking across the equal-MI null. The larger-sample 4×4 comparison has practically small differences under both calibrations.

The method does not resolve errors in the shared statistic. In sparse different-skew tables, MI bias can shift or reverse the estimated difference and cause rejection to fall initially as the true difference increases. Same-skew populations can instead remain conservative when the expected difference bias cancels. The matched construction and allocation checks show why population structure and sample allocation matter alongside the cutoff. Expanded Welch therefore offers useful behaviour for some categorical comparisons, while accurate estimation of the MI difference and its uncertainty remains necessary for reliable inference.

8.2 Limitations and future work

The findings concern fixed categorical populations, the tested association constructions and the specified alternative paths. They support further investigation of the MI-specific rule rather than its use as a general replacement for Normal Wald. The independently fitted references and calibration thresholds assess mechanisms under known populations; they do not supply an immediately available procedure for arbitrary observed tables.

A first extension is to derive the second-order mean contribution to the variance estimator and evaluate a corresponding degrees-of-freedom correction. A second is to improve sparse MI bias and standard-error estimation, assessing their effects separately through null rejection, power curves and estimated direction.

Near independence, the quadratic sampling approximation may support a more accurate comparison of MI differences. Any likelihood-ratio or resampling procedure must preserve the equal-MI null even when the complete population tables differ. Reversed power paths, stronger dependence at small sample sizes, additional association constructions and broader allocation comparisons would assess whether power rankings are stable across a wider range of equal-MI populations.

Appendix A

Supporting Calculations

These calculations support the variance sensitivity, near-independence interpretation and validity conditions in Chapter 4.

A.1 Mean of the variance influence function

The observation-level sensitivities $g_P(x, y)$ have a probability-weighted mean of zero. Averaging the one-cell perturbation directions leaves the population table unchanged:

$$\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = P - P = 0. \tag{A.1}$$

This weighted cancellation also holds after exact plug-in substitution. Because the first derivative is linear in the direction of change, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$\begin{aligned} \mathrm{E}_P\{g_P(X, Y)\} &= \sum_{x,y} p_{xy} \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\{(1-\varepsilon)P + \varepsilon\delta_{(x,y)}\}\Big|_0 && \text{(A.2)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\}\right\}\Big|_0 && \text{(A.3)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V(P)\Big|_0 = 0. && \text{(A.4)} \end{aligned}$$

A.2 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t, all cells remain positive and the margins remain a and b. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.5}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i,j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.6}$$

using $\log(1 + z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.7}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.8}$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned} \mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X = i\} &= \frac{1}{a_i} \sum_j (q_{ij} + tH_{ij}) \left\{ \frac{tH_{ij}}{q_{ij}} + O(t^2) \right\} \\ &= \frac{t}{a_i} \sum_j H_{ij} + O(t^2) = O(t^2). \end{aligned}$$

The last equality uses the zero row sum of H. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.29), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order t^2. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.29) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.9}$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned} \tau^2(P_t) &= \sum_{i,j} (q_{ij} + tH_{ij}) \left\{ \frac{2tH_{ij}}{q_{ij}} + O(t^2) \right\}^2 \\ &= 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \end{aligned} \tag{A.10}$$

Comparing the leading terms of $I(P_t)$, $V(P_t)$ and $\tau^2(P_t)$ gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Appendix A.4 treats that case separately.

A.3 Dependence between the MI and variance estimates

Both $\widehat{I}(P)$ and $\widehat{V}(P)$ are calculated from the same sample. Under fixed positive support, their first-order covariance is

$$\mathrm{Cov}\{\widehat{I}(P), \widehat{V}(P)\} \approx \frac{1}{n_P} \mathrm{E}_P[\{\ell_P(X,Y) - I(P)\} g_P(X,Y)]\,. \tag{A.11}$$

The corresponding expression for Q replaces P and n_P with Q and n_Q. To determine how the MI and variance estimates move together near independence, use the fixed-margin path $P_t = (q_{ij} + tH_{ij})$ from Appendix A.2. For small positive t, the expansions there give

$$\ell_{P_t}(i,j) - I(P_t) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \qquad g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.12}$$

Substituting both expressions into the first-order covariance above gives

$$\mathrm{Cov}\{\widehat{I}(P_t), \widehat{V}(P_t)\} \approx \frac{2t^2}{n_P} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O\left(\frac{t^3}{n_P}\right) \tag{A.13}$$

$$= \frac{2V(P_t)}{n_P} + O\left(\frac{t^3}{n_P}\right). \tag{A.14}$$

The first-order covariance is therefore positive for small positive MI along this path: samples with a larger MI estimate also tend to have a larger variance estimate. In the two-sample statistic, P's and Q's MI estimates enter the numerator with opposite signs, whereas their variance contributions add under the square root in the standard error. At exact independence, $V(P_0) = 0$ and a second-order sampling approximation is needed.

A.4 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table P and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.15}$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.16}$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.17}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.18}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.19}$$

The first sum measures squared cell changes, whereas the last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \widehat{P} - P$. Since cell-probability errors typically have size proportional to $n^{-1/2}$, squaring them makes $\widehat{I}(P)$ proportional to n^{-1} under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.20}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

Sampling variation at small positive MI.

The quadratic approximation also explains why estimated degrees of freedom remain several units when population MI is almost zero. Keep r and c fixed and let a positive population table approach independence with cell departures proportional to $n^{-1/2}$. The normal sampling errors and the population departure then contribute on the same scale. In regular models, the likelihood-ratio statistic has a noncentral chi-squared limit under local alternatives (Banerjee, 2005). For the multinomial model

here, assume strictly positive limiting cell probabilities. The local approximation is

$$G = 2n\widehat{I}(P) \mathrel{\dot\sim} \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \simeq 2nI(P), \tag{A.21}$$

where $\mathrel{\dot\sim}$ denotes an approximate distribution. The noncentral chi-squared distribution describes a sum of squared normal errors with a nonzero mean departure; κ measures that departure. At independence, $\kappa = 0$, recovering Equation (A.20).

For a sampled table close to independence, the expansions in Appendix A.2, applied around its own independent margins, give

$$\widehat{V}(P) \approx 2\widehat{I}(P) = \frac{G}{n}, \qquad \widehat{\tau}^2(P) \approx 4\widehat{V}(P) \approx \frac{4G}{n}. \tag{A.22}$$

Substitution into the implemented component formula gives

$$\widehat{\nu}_P \approx \frac{2n(G/n)^2}{4G/n} = \frac{G}{2}. \tag{A.23}$$

Thus sampling noise contributes to the estimated degrees of freedom even when the population value $nI(P)$ is very small.

The same approximation separates that plug-in calculation from matching the actual sampling moments. Since $\mathrm{E}(G) = d + \kappa$ and $\mathrm{Var}(G) = 2(d + 2\kappa)$,

$$\mathrm{E}\{\widehat{V}(P)\} \approx \frac{d+\kappa}{n}, \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{2(d+2\kappa)}{n^2}. \tag{A.24}$$

Keeping both terms in the moment-matching equation gives

$$\nu_P^{\mathrm{mom}} \approx \frac{2\{(d+\kappa)/n\}^2}{2(d+2\kappa)/n^2} = \frac{(d+\kappa)^2}{d+2\kappa}. \tag{A.25}$$

When κ is much larger than d, this reduces to $\kappa/2 \simeq nI(P)$, as in Equation (4.47). When κ is small, however, the matched degrees of freedom approach d, while the mean plug-in degrees of freedom approach $d/2$.

For the 4×4, $n = 1{,}000$, $I(P) = 0.0001$ setting, $d = 9$ and $\kappa \simeq 0.2$. The local calculation gives a mean estimated component degree of freedom of approximately 4.6, rather than the population first-order value 0.1. It also predicts that the average plug-in estimate of $\mathrm{Var}\{\widehat{V}(P)\}$ divided by its actual sampling variance is approximately

$$\frac{4(d+\kappa)/n^2}{2(d+2\kappa)/n^2} = \frac{2(d+\kappa)}{d+2\kappa} \simeq 1.96. \tag{A.26}$$

This explains the scale of the estimated degrees of freedom and the roughly twofold plug-in overprediction in Table 6.1. The population formula alone omits the quadratic sampling contribution responsible for both effects.

A.5 Combined degrees of freedom with a zero variance estimate

For positive component degrees of freedom, substituting Equation (4.39) and its Q analogue into Equation (4.40) gives the equivalent combined form

$$\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\widehat{\tau}^2(P)/(2n_P^3) + \widehat{\tau}^2(Q)/(2n_Q^3)}. \tag{A.27}$$

This direct form remains defined when one table has $\widehat{V} = \widehat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. The implementation nevertheless requires both components to have positive, finite degrees of freedom, so it records this case as invalid.

A.6 Numerical derivative checks

The analytic derivatives in Equations (4.18) and (4.29) were checked by centred finite differences along the one-cell perturbation paths in Chapter 4. Two positive tables,

$$\begin{pmatrix} 0.36 & 0.14 \\ 0.09 & 0.41 \end{pmatrix}, \qquad \begin{pmatrix} 0.28 & 0.09 & 0.07 \\ 0.08 & 0.17 & 0.06 \\ 0.04 & 0.10 & 0.11 \end{pmatrix},$$

were checked at every cell with step size $\varepsilon = 10^{-6}$. For each path, the difference between the values at $+\varepsilon$ and $-\varepsilon$ was divided by 2ε; all perturbed probabilities remained positive. The maximum absolute error was of order 10^{-10} for both derivatives.

Appendix B

Experimental Details

This appendix specifies the population formulas, target-MI search, complete parameter grids, sampling procedure and outcome calculations. Together with the test formulas in Chapters 3–5, these details define the statistical experiment.

B.1 Copying construction and target selection

For a positive marginal vector a, the joint distribution is

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}. \tag{B.1}$$

Summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.2}$$

Summing a column similarly gives a_j. The copying branch contributes only to its matching category, while the independent branch contributes across every category. Thus the construction preserves both margins and has positive cells whenever $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of a. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.3}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes

the first derivative positive for $\lambda > 0$. At the endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.4}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are 10^{-15} and 10^{-14}. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive, its total and marginal errors are at most 10^{-12}, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. Every target is checked against its own population entropy endpoint, including the alternatives scaled by the fitted null standard deviation.

B.2 Four-cell supporting construction

For the supporting 4×4 and 16×16 comparisons, begin with the independence probabilities $a_i a_j$ and transfer an amount t within the first two rows and columns. The resulting table $R(t)$ has cell probabilities

$$r_{ij}(t) = \begin{cases} a_i a_j + t, & (i, j) = (1, 1) \text{ or } (2, 2), \\ a_i a_j - t, & (i, j) = (1, 2) \text{ or } (2, 1), \\ a_i a_j, & \text{all other cells.} \end{cases} \tag{B.5}$$

Every affected row and column gains and loses t, preserving its total. The decreasing cells have starting probability $a_1 a_2$, so

$$0 \le t < t_{\max} = a_1 a_2. \tag{B.6}$$

Solve $I\{R(t)\} = I_{\text{target}}$ by the same root search on $[0, t_{\max}]$. The target must lie below the endpoint MI, and the constructed table passes the same checks as the copying table.

In a binary table the copying construction is exactly this transfer:

$$P(\lambda) = (1 - \lambda) \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix} \tag{B.7}$$

$$= \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda a_1 a_2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{B.8}$$

The second equality uses $a_1 + a_2 = 1$, giving $a_1 - a_1^2 = a_1 a_2$ and $a_2 - a_2^2 = a_1 a_2$. Hence $t = \lambda a_1 a_2$. This equivalence does not extend to the larger tables: copying changes cells throughout the table, whereas Equation (B.5) changes only four.

B.3 Complete parameter grids and sampling

Table B.1 lists the parameter combinations for each study and supporting check. Table sizes are $k \in \{2, 4, 8, 16\}$. Each population has identical row and column margins. Uniform margins give probability $1/k$ to every category; same skew uses $m_k(0.8)$ in both populations; different skew uses $m_k(0.7)$ in P and $m_k(0.8)$ in Q, with m_k defined by Equation (5.3). The same copying formula and root search apply to every primary-study setting.

Table B.1: Exact parameter sets; MI values are in nats.

Study	Settings
Equal MI	All four sizes and three margin regimes; $I(P) = I(Q) = 0.02$; equal sample sizes $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Population MI	4×4, uniform, same skew and different skew; $I(P) = I(Q) \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05\}$; $n \in \{100, 1000, 10000\}$.
Power	All four sizes with same skew and different skew, plus uniform 4×4; $I(P) = 0.02$, $I(Q) = 0.02 + \lvert\Delta\rvert$; $\lvert\Delta\rvert \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$; $n \in \{1000, 10000\}$.
Focused power	Settings A–C in Table 5.3; $I(P) = 0.02$, $I(Q) = 0.02 + h\sigma_0$, $h \in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}$. The fitted null SDs defining these populations are given below.
Calibration sensitivity	Settings A–C at $h \in \{2, 4\}$; upper null $I(P) = I(Q) = 0.02 + h\sigma_0$, retaining each setting's margins, sample sizes and copying construction. Both calibrations are applied to the fixed alternatives in the focused-power row.

Study	Settings
Four-cell check	$k \in \{4, 16\}$, same skew and different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $n \in \{1000, 10000\}$; construction in Equation (B.5).
Sample allocation	Copying 4×4, same skew and different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $(n_P, n_Q) \in \{(1000, 1000), (400, 1600), (1600, 400)\}$. The total sample size is 2,000 in every comparison.

For the focused power comparison, let $\sigma_0 = \mathrm{SD}_0(\widehat{\Delta}_{\mathrm{BC}})$ be the SD estimated from the independent null fitting batch, centring at its sampling mean. The grid is

$$I(P) = 0.02, \qquad I(Q) = 0.02 + h\sigma_0, \qquad h \in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}. \tag{B.9}$$

The true signed difference is $\Delta = -h\sigma_0$. Population tables and differences in nats are fixed after this construction.

The fixed null-SD constants used to construct the focused alternatives are

$$\begin{aligned} \sigma_{0,A} &= 0.154966805570126 \text{ nats}, \\ \sigma_{0,B} &= 0.065137776499116 \text{ nats}, \\ \sigma_{0,C} &= 0.003174953145465 \text{ nats}. \end{aligned}$$

For each h in the focused grid, set the target MI of Q to $0.02 + h\sigma_0$ and solve for its copying probability. The fitting uncertainty of these constants is reported in Table C.3; the constructed populations remain fixed during evaluation.

For one repetition, draw

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \mathrm{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.10}$$

and independently draw the corresponding Q counts. Reshape these counts into their $k \times k$ tables and apply both tests and the fixed moment reference to that pair. Repeat independently 100,000 times per distinct evaluation setting. Independently generate another 100,000 pairs for each equal-MI fitting setting. The simulations use the PCG64 pseudo-random generator with master seed 20261021 and separate streams for population, setting, fitting and evaluation samples. Repeated settings use one set of repetitions in every relevant display.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms, $0 \log 0 = 0$, and no smoothing. The bias correction for each plug-in MI is $(k-1)^2/(2n)$; do not truncate negative corrected estimates. In the equal-sample studies, the two corrections cancel in their difference. In the allocation check, Equation (3.7) subtracts $9/(2n_P)$ and adds $9/(2n_Q)$ to the plug-in difference. Their combined contribution is -0.0084375 nats at $(400, 1600)$ and $+0.0084375$ at $(1600, 400)$. These terms remove the leading unequal-sample bias; the diagnostics measure the remaining bias. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

B.4 Validity and sparse-sample measures

Normal Wald requires a finite corrected difference, a finite positive standard error, $\widehat{V}(P)+\widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. Expanded Welch additionally requires positive finite degrees of freedom for both components and their combination, and a finite expanded p-value. A failed component makes Expanded Welch invalid under this rule, including the zero-component case in Appendix A.5.

The fixed reference requires the base conditions and a positive finite fitted reference value. No component-degree requirement is added to the fixed rule.

The primary rate divides valid rejections by all repetitions. A method-valid rate uses only that method's valid repetitions; a common-valid rate uses only pairs where both tests and the fixed reference are valid. Report the denominator for either conditional rate, which is undefined when the denominator is zero. Invalid outputs contribute no rejection to the primary rate.

Record the number of zero-count cells, empty rows and empty columns separately for each population. These are sampling diagnostics rather than eligibility rules. The minimum expected cell count is $\min_{ij} np_{ij}$, with the corresponding definition for Q. The study applies no minimum expected-count filter.

B.5 Bias, variance and degrees-of-freedom diagnostics

Let $\widehat{\Delta}_s$ be the corrected difference in repetition s. Its bias estimate is the mean of the finite-valued MI difference estimates minus the known Δ. Its observed sampling

variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.11}$$

where M counts those finite-valued estimates. Centre at their sample mean, rather than the true difference, so this measures variance separately from bias. Include finite-valued MI differences even when the test output is invalid. For P and Q, the individual corrected MI biases are the mean finite corrected MI estimates minus $I(P)$ and $I(Q)$, respectively.

For the estimated/observed variance ratio, restrict both averages to the same repetitions with a finite difference and finite squared standard error. Retain finite zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset; the ratio is undefined if the observed variance is zero. Report its included fraction. The population comparison instead uses $V(P)/n_P + V(Q)/n_Q$ and the primary observed variance.

The Monte Carlo standard error of estimated bias is the observed SD divided by $\sqrt{M}$. Component degrees-of-freedom summaries use positive finite component values, with their included fraction reported. Combined summaries use positive finite combined values. Finite zero variance estimates remain in the moment calculations even when component degrees of freedom are undefined. Percentile ranges describe the spread of those outputs, not confidence intervals for their median. These summaries expose estimation problems but do not isolate the causal contribution of each problem to rejection. The figures show a mean-estimate-to-moment-target ratio only where all component degrees of freedom are defined. Conditional means elsewhere retain their inclusion counts in the numerical summaries.

The dependence diagnostic uses Spearman's rank correlation between $|\widehat{\Delta}_s - \overline{\widehat{\Delta}_{\mathrm{BC}}}|$ and $\widehat{\mathrm{SE}}_s$, where the mean is taken over repetitions with both quantities finite. No validity restriction removes finite pairs; included and excluded counts accompany the summaries. Centring removes the mean shift so the diagnostic measures whether larger deviations accompany larger estimated standard errors. With identical populations and equal sample sizes, exchanging the two samples reverses the signed difference but leaves the standard error unchanged. A zero signed correlation can therefore occur by symmetry even when their magnitudes are dependent.

Variability of the estimated MI variance.

For each selected population, measure the observed variability of $\widehat{V}(P)$ by replacing $\widehat{\Delta}_s$ with $\widehat{V}_s(P)$ in Equation (B.11). This measures how much the variance estimate itself changes between samples. Chapter 4 predicts that variability to first order.

Compare the observed value with

$$\begin{aligned} \text{population prediction} &= \frac{\tau^2(P)}{n_P}, \\ \text{mean plug-in prediction} &= \frac{1}{M}\sum_{s=1}^{M}\frac{\widehat{\tau}_s^2(P)}{n_P}. \end{aligned} \tag{B.12}$$

The population prediction uses the known probabilities, whereas the mean plug-in prediction uses the empirical probabilities in each repetition. Repeat both for Q. For each population, use the same repetitions with finite $\widehat{V}_s$ and $\widehat{\tau}_s^2$, including finite zeros and outputs whose test is invalid. Divide each prediction by the observed variability: a ratio of one means agreement, and the ratio is undefined when observed variability is zero.

Estimated direction among rejections.

For a nonzero population difference, classify each valid two-sided rejection by the sign of $\widehat{\Delta}_{\mathrm{BC}}$. The estimate has the opposite sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta < 0$; it has the correct sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta > 0$. Report the number of rejections and the number with opposite signs for each method. The fraction with opposite signs divides the opposite-sign count by the rejection count and is undefined if there are no rejections. These counts supplement the estimate of power, which includes every valid rejection regardless of direction.

B.6 Simulation uncertainty

For a rate estimate $\widehat{r}$ from M independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Wilson intervals use $z = \Phi^{-1}(0.975)$ and have centre and half-width

$$\frac{\widehat{r} + z^2/(2M)}{1 + z^2/M}, \qquad \frac{z\sqrt{\widehat{r}(1-\widehat{r})/M + z^2/(4M^2)}}{1 + z^2/M}. \tag{B.13}$$

These intervals are used for both rejection and validity rates (Wilson, 1927).

For the paired method contrast, let D_s equal the Expanded Welch valid-rejection indicator minus the comparator valid-rejection indicator on pair s. The approximate interval for the average paired difference $\bar{D}$ is

$$\bar{D} \;\pm\; 1.96\left\{\frac{1}{M(M-1)}\sum_{s=1}^{M}(D_s - \bar{D})^2\right\}^{1/2}. \tag{B.14}$$

Use the same common-valid subset when reporting a conditional paired contrast. Cross-construction comparisons use independent simulation streams, as do different sample allocations, so their difference uncertainty is not calculated from paired indicators. All intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

B.7 Moment targets and the fixed reference

The first comparison in Chapter 5 measures the mean and sampling variance of $\widehat{V}(P)$ separately. Expectations and variances refer to repeated samples from the same fixed population at the same sample size. Substituting the predictions in Equation (5.4) into the moment-matching formula gives

$$\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)}. \tag{B.15}$$

The subscript FO identifies the population first-order formula from Chapter 4. The sampling-moment target instead uses the actual mean and variance of the estimator:

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{B.16}$$

The assumptions in Chapter 4 ensure positive population variance quantities for the tested populations. The two degrees-of-freedom values can nevertheless differ at finite sample sizes. Dividing their definitions before applying either approximation gives the exact identity

$$\frac{\nu_{P,\mathrm{FO}}}{\nu_{P,\mathrm{mom}}} = \left\{\frac{V(P)}{\mathrm{E}\,\widehat{V}(P)}\right\}^2 \frac{\mathrm{Var}\{\widehat{V}(P)\}}{\tau^2(P)/n_P}. \tag{B.17}$$

The first factor measures the contribution of the mean approximation; the second measures the contribution of the sampling-variance approximation. A mean estimate above $V(P)$ lowers the first-order degrees of freedom relative to the moment target when the variance prediction is otherwise accurate. The same calculations are made separately for Q.

The fixed diagnostic rule uses a moment-matched reference, estimated from independent simulations of the corresponding equal-MI populations:

$$\nu_{\mathrm{ref}} = \frac{2\{\mathrm{E}(\widehat{\mathrm{SE}}^2)\}^2}{\mathrm{Var}(\widehat{\mathrm{SE}}^2)}. \tag{B.18}$$

Independence of the population samples gives

$$\mathrm{E}(\widehat{\mathrm{SE}}^2) = \frac{\mathrm{E}\,\widehat{V}(P)}{n_P} + \frac{\mathrm{E}\,\widehat{V}(Q)}{n_Q}, \tag{B.19}$$

$$\mathrm{Var}(\widehat{\mathrm{SE}}^2) = \frac{\mathrm{Var}\{\widehat{V}(P)\}}{n_P^2} + \frac{\mathrm{Var}\{\widehat{V}(Q)\}}{n_Q^2}. \tag{B.20}$$

Substitution of the component moment targets into the combination formula at these expected variance contributions gives exactly Equation (B.18). Thus the combination identity introduces no further approximation at the level of moments. The next section gives the cancellation explicitly. Independent fitting samples estimate these moments by their sample means and sample variances. The Student t distribution remains an approximation to the shared statistic's distribution.

The fixed reference uses the two-sided p-value formula in Equation (4.41), substituting its fitted degrees of freedom. A valid p-value at most 0.05 produces a nominal rejection.

The moment reference requires known simulation populations and is held fixed within an evaluation setting and across the alternatives belonging to the same null. For an alternative, the fitting populations have equal MI of 0.02 nats and the same table size, margins, construction and sample allocation as that alternative. Expanded Welch versus this reference measures the change in rejection when the sample-dependent degrees of freedom are replaced by that fixed value. The reference can still reject incorrectly when the shared difference is biased, its standard error has the wrong scale, or their joint distribution is poorly described by the Student t distribution.

B.8 Combining component moment targets

The combined moment reference follows exactly from the component moments when the two samples are independent. This calculation assumes positive component means, finite component variances and a positive variance of the summed contributions. Substitute the expected variance contributions and component moment degrees of freedom into the combination formula:

$$\frac{\{\mathrm{E}\,\widehat{V}(P)/n_P + \mathrm{E}\,\widehat{V}(Q)/n_Q\}^2}{\dfrac{\{\mathrm{E}\,\widehat{V}(P)/n_P\}^2}{2\{\mathrm{E}\,\widehat{V}(P)\}^2/\,\mathrm{Var}\{\widehat{V}(P)\}} + \dfrac{\{\mathrm{E}\,\widehat{V}(Q)/n_Q\}^2}{2\{\mathrm{E}\,\widehat{V}(Q)\}^2/\,\mathrm{Var}\{\widehat{V}(Q)\}}}. \tag{B.21}$$

For either component, the square of the mean cancels:

$$\frac{\{\mathrm{E}\,\widehat{V}(P)/n_P\}^2}{2\{\mathrm{E}\,\widehat{V}(P)\}^2/\operatorname{Var}\{\widehat{V}(P)\}} = \frac{\{\mathrm{E}\,\widehat{V}(P)\}^2}{n_P^2}\frac{\operatorname{Var}\{\widehat{V}(P)\}}{2\{\mathrm{E}\,\widehat{V}(P)\}^2} = \frac{\operatorname{Var}\{\widehat{V}(P)\}}{2n_P^2}. \tag{B.22}$$

Applying the same cancellation to Q reduces Equation (B.21) to

$$\frac{2\{\mathrm{E}\,\widehat{V}(P)/n_P + \mathrm{E}\,\widehat{V}(Q)/n_Q\}^2}{\operatorname{Var}\{\widehat{V}(P)\}/n_P^2 + \operatorname{Var}\{\widehat{V}(Q)\}/n_Q^2} = \frac{2\{\mathrm{E}(\widehat{\mathrm{SE}}^2)\}^2}{\operatorname{Var}(\widehat{\mathrm{SE}}^2)}. \tag{B.23}$$

The final equality uses independence to remove the covariance between $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. Thus matching each component and combining at its expected contribution yields the same target as matching the sum directly. This identity concerns moments; it does not assert that the sum has a scaled chi-squared distribution. Using random sample contributions instead of their expectations produces a different, sample-dependent degrees-of-freedom rule.

B.9 Moment estimation and Monte Carlo uncertainty

For a component, let $X_s = \widehat{V}_s$ for repetition s, with population sampling mean $\mu = \mathrm{E}(X_s)$ and sampling variance $\sigma^2 = \operatorname{Var}(X_s) > 0$. From M repetitions, calculate

$$\bar{X} = \frac{1}{M}\sum_{s=1}^{M} X_s, \qquad S_X^2 = \frac{1}{M-1}\sum_{s=1}^{M}(X_s - \bar{X})^2. \tag{B.24}$$

These moments retain finite zero values. Estimate the moment degrees of freedom by $2\bar{X}^2/S_X^2$. The diagnostic decomposition uses these evaluation moments, whereas the reference cutoff uses the independent fitting moments. An undefined component degree of freedom does not remove its finite variance estimate from either calculation.

To quantify uncertainty in the moment degrees of freedom, apply the multivariate delta method to $f(m, v) = 2m^2/v$. The first-order Taylor formula for a function of two arguments is

$$f(m + \Delta m, v + \Delta v) \approx f(m, v) + \frac{\partial f}{\partial m}(m, v)\Delta m + \frac{\partial f}{\partial v}(m, v)\Delta v. \tag{B.25}$$

Here the expansion point is (μ, σ^2), the estimated point is $(\bar{X}, S_X^2)$, and the differences are $\bar{X} - \mu$ and $S_X^2 - \sigma^2$. The derivatives are

$$\frac{\partial f}{\partial m} = \frac{4m}{v}, \qquad \frac{\partial f}{\partial v} = -\frac{2m^2}{v^2}. \tag{B.26}$$

Substitution gives

$$\frac{2\bar{X}^2}{S_X^2} - \frac{2\mu^2}{\sigma^2} \approx \frac{4\mu}{\sigma^2}(\bar{X} - \mu) - \frac{2\mu^2}{(\sigma^2)^2}(S_X^2 - \sigma^2). \tag{B.27}$$

The mean and variance estimation errors come from the same repetitions, so their covariance must be retained. Their first-order contributions for one repetition are estimated by

$$U_s = X_s - \bar{X}, \qquad W_s = (X_s - \bar{X})^2 - S_X^2. \tag{B.28}$$

Using the fitted derivatives, the contribution for the degrees-of-freedom estimate is

$$Z_s = \frac{4\bar{X}}{S_X^2} U_s - \frac{2\bar{X}^2}{(S_X^2)^2} W_s. \tag{B.29}$$

The sample SD of Z_s, divided by $\sqrt{M}$, estimates the Monte Carlo standard error. Taking the variance of the combined contribution preserves the covariance between the estimated mean and variance. The combined reference uses the corresponding weighted contributions from both independent populations. Mean and variance ratios use the same delta-method calculation; the plug-in prediction and observed variance are kept paired when both are estimated from the same outputs.

Nominal reference-rate intervals condition on the fitted degrees of freedom. Its fitting Monte Carlo standard error is reported separately. For the calibrated comparison, refitting the fixed degrees of freedom would not change the ordering by $|T|$; the calibration bootstrap therefore resamples its empirical absolute-statistic cutoff directly.

B.10 Empirical calibration and power uncertainty

For a sample-dependent rule, assign an invalid output p-value one for calibration and order the fitting p-values. The threshold is the $(\lfloor 0.05M \rfloor + 1)$th smallest value, and rejection requires a strictly smaller evaluation p-value. Every tie at the boundary is excluded. Normal Wald and the fixed reference instead share the equivalent empirical upper-tail threshold for $|T|$, with invalid fitting values placed below all valid absolute statistics. This convention gives at most 5% rejection in the fitting

batch. Independent null evaluation measures the achieved rate, including any effect of discreteness.

For focused calibrated power, resample the fitting outputs and refit these thresholds in each of 1,000 bootstrap repetitions. Apply a common resampled fitting batch across all points of a curve. At each evaluation point, resample the joint decisions of both tests and the fixed moment reference, preserving their pairing. Use the 2.5th and 97.5th percentiles for pointwise intervals of rates and paired rate differences. Compare the paired intervals with the practical tolerance of minus one to plus one percentage point. The fixed reference and Normal Wald remain identical after this calibration, so refitting their moment degrees of freedom cannot change the calibrated decisions. The population tables and effect sizes in nats stay fixed throughout the bootstrap.

B.10.1 Sensitivity to the calibration population

For each focus setting and $h = 2, 4$, retain the alternative $(I(P), I(Q)) = (0.02, 0.02 + h\sigma_0)$. The lower fitting null is $(0.02, 0.02)$; construct the upper null by setting both MI values to $0.02 + h\sigma_0$. For each population, solve for its copying probability using its original margin vector, retaining the dimensions and sample sizes in Table 5.3.

Generate 100,000 new fitting pairs and 100,000 independent validation pairs for each of these six upper nulls. Compare lower- and upper-null calibration on the same original alternative pairs, then apply both sets of cutoffs to the original lower-null and new upper-null validation pairs. Invalid outputs remain non-rejections in every comparison. Validation concerns these two null populations.

For uncertainty in power and its contrasts, use 1,000 joint bootstrap repetitions. In each repetition, resample the lower fitting pairs once per focus setting and share that resample across its two alternatives. Independently resample each upper fitting batch and refit all cutoffs. For a fixed alternative, jointly resample the decisions of both tests and the fixed moment reference under both calibrations. This retains pairing between rules and between calibrations when estimating gains and changes in gains. Use percentile 95% intervals, keeping population tables and effect sizes fixed. Cross-null rate intervals use the Wilson calculation conditional on the fitted cutoffs; they quantify validation sampling uncertainty only.

Appendix C

Additional Results

This appendix reports the component diagnostics, calibration sensitivity, rejection directions and multiple-testing adjustment used in the main argument. Each simulation setting uses 100,000 evaluation pairs. The experimental supplement retains the full companion plots and supporting comparisons.

C.1 Component diagnostics and reference fitting

The mean error of the variance estimator varies with population MI and margins. For the 4×4 sweep, the scaled error $n\{\mathrm{E}\,\widehat{V} - V\}$ ranges from 5.55 to 8.45 at $n = 100$, 7.50 to 9.39 at $n = 1{,}000$, and 7.60 to 10.19 at $n = 10{,}000$. The coefficient suggested near independence is $(k-1)^2 = 9$. Variation in the observed scaled error leaves residual mean error under a fixed $9/n$ correction.

Table C.1 gives the means, medians and fifth–95th percentile ranges of component degrees of freedom. The moment target uses the evaluation sampling moments. All 100,000 component estimates are positive and finite in each row; percentile ranges describe sampling spread.

Table C.2 describes the shared difference and standard error. Bias/SD divides difference bias by its observed SD; variance ratio divides the mean squared standard error by the observed difference variance. Rank corr. is Spearman's correlation between the absolute centred difference and its standard error, and skew describes the difference estimates. All 100,000 pairs are included in each row. A near-zero signed correlation can follow from symmetry despite magnitude dependence: the same-skew 2×2, $n = 100$ null has signed correlation about -0.001 and magnitude rank correlation 0.547.

Table C.1: Sampling spread of component degrees of freedom.

Setting, pop.	Mean df	Median df	5th–95th	Moment df	Included
A, P	6.52	5.23	[1.95, 15.32]	14.81	100,000
A, Q	6.54	5.24	[1.93, 15.16]	14.87	100,000
B, P	14.04	12.74	[5.83, 26.60]	23.54	100,000
B, Q	9.72	8.47	[2.64, 20.91]	12.47	100,000
C, P	157.09	156.66	[131.26, 184.27]	157.96	100,000
C, Q	139.70	139.28	[115.88, 165.12]	141.27	100,000
16, same, 1,000, P	39.78	39.32	[29.18, 51.92]	71.67	100,000
16, same, 1,000, Q	39.80	39.35	[29.33, 51.88]	72.18	100,000
16, same, 10,000, P	133.18	132.82	[114.30, 153.25]	148.48	100,000
16, same, 10,000, Q	133.17	132.92	[114.19, 153.21]	146.84	100,000
16, different, 1,000, P	52.58	52.08	[41.98, 64.93]	87.28	100,000
16, different, 1,000, Q	39.80	39.33	[29.15, 51.94]	71.05	100,000
16, different, 10,000, P	172.67	172.41	[152.55, 193.72]	189.98	100,000
16, different, 10,000, Q	133.24	132.99	[114.21, 153.33]	147.07	100,000

Table C.2: Shared-estimator diagnostics and combined degrees of freedom.

Setting	Bias/SD	Var. ratio	Rank corr.	Skew	Median df	Ref. df
A	0.003	0.827	0.017	0.002	10.61	29.70
B	0.870	1.535	0.122	-0.052	21.12	33.34
C	0.000	1.010	0.060	-0.013	293.23	293.98
16, same, 1,000	-0.003	1.929	0.106	0.008	78.87	142.90
16, same, 10,000	0.001	1.047	0.066	0.012	265.85	294.14
16, different, 1,000	2.058	1.938	0.114	-0.008	88.51	155.70
16, different, 10,000	-0.229	1.064	0.055	-0.013	299.21	329.92

The fixed references and null SDs in Table C.3 use independent 100,000-pair fitting batches. Their Monte Carlo standard errors follow Appendix B.9. The fitted SDs define the fixed alternative populations, so power intervals condition on those tables.

Table C.3: Independent moment references and fitted null standard deviations.

Setting	Reference df	MCSE	Null SD (nats)	MCSE (nats)
A	29.70	0.10	0.154967	0.000363
B	33.34	0.11	0.065138	0.000152
C	293.98	0.94	0.003175	0.000007

C.2 Calibrated power and sensitivity checks

Figure C.1 shows the full power curves under calibration at common MI 0.02. Wald and the fixed moment reference have identical calibrated decisions. Table C.4 gives nominal and calibrated paired contrasts at two null SDs, including cutoff-fitting uncertainty for the calibrated intervals.

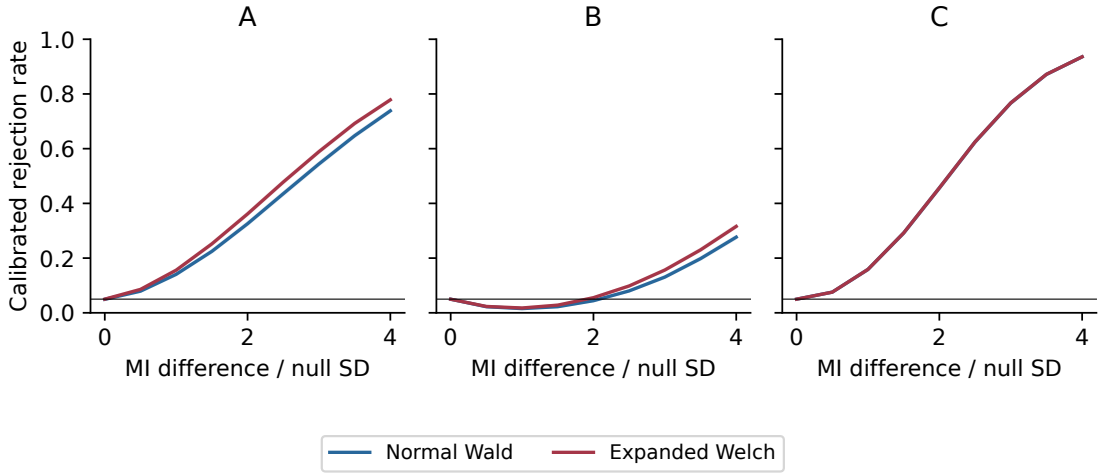


Figure C.1: Power after calibration at common MI 0.02; bands show bootstrap uncertainty.

Table C.5 compares both calibrations on the same fixed alternatives. The lower null has common MI 0.02; the upper null sets both MI values to the alternative's larger value. Appendix B.10.1 specifies the construction and joint bootstrap. Table 6.3 in Chapter 6 gives the Wald contrasts with intervals.

Table C.4: Expanded Welch minus Normal Wald at two null SDs, in percentage points.

Setting	Nominal	95% interval	Calibrated	95% interval
A	-5.128	[-5.265, -4.991]	+3.580	[+3.295, +3.787]
B	-0.727	[-0.780, -0.674]	+1.112	[+1.028, +1.194]
C	-0.304	[-0.338, -0.270]	+0.044	[+0.013, +0.071]

Table C.5: Power for fixed alternatives under lower- and upper-null calibration.

Setting: SD, null	Wald/ref.	Expanded
A: 2, lower	32.57%	36.15%
A: 2, upper	18.58%	17.83%
A: 4, lower	73.84%	77.84%
A: 4, upper	54.91%	55.09%
B: 2, lower	4.42%	5.54%
B: 2, upper	4.68%	4.73%
B: 4, lower	27.66%	31.59%
B: 4, upper	26.40%	26.01%
C: 2, lower	45.61%	45.66%
C: 2, upper	45.47%	45.42%
C: 4, lower	93.56%	93.58%
C: 4, upper	93.42%	93.41%

Table C.6 applies each fitted cutoff to both null validation populations. Each row includes all 100,000 pairs, with invalid outputs contributing no rejection. The two null populations check sensitivity to calibration; they do not establish control across the complete equal-MI null.

Table C.6: Independent null validation under both fitted calibrations.

		Lower evaluation null		Upper evaluation null	
Setting: SD	Fitting null	Wald/ref.	Expanded	Wald/ref.	Expanded
A: 2	lower	4.93%	4.99%	11.20%	13.71%
A: 2	upper	1.63%	0.79%	4.87%	4.84%
A: 4	lower	4.93%	4.99%	12.56%	15.58%
A: 4	upper	1.34%	0.53%	5.04%	5.00%
B: 2	lower	4.98%	4.96%	4.82%	5.76%
B: 2	upper	5.26%	4.16%	5.04%	5.04%
B: 4	lower	4.98%	4.96%	5.37%	6.76%
B: 4	upper	4.56%	3.27%	4.99%	4.99%
C: 2	lower	4.98%	4.98%	4.97%	4.99%
C: 2	upper	4.93%	4.90%	4.94%	4.94%
C: 4	lower	4.98%	4.98%	5.03%	5.06%
C: 4	upper	4.84%	4.81%	4.87%	4.86%

C.3 Sparse-table bias and estimated direction

For 16×16 different-skew populations at $n = 1{,}000$, the true difference is $I(P) - I(Q) = -0.01$ nats. In this setting, 10,183 of 10,184 Wald rejections and 9,591 of 9,592 Expanded Welch rejections have the wrong estimated ordering. The fixed reference gives 9,829 wrong-direction cases among 9,830 rejections. Under same skew, the corresponding Wald and Expanded Welch counts are 59 of 1,809 and 52 of 1,646. At $n = 10{,}000$, neither test nor the fixed reference has a wrong-direction rejection at this alternative under either skew regime. Under different skew, the difference bias at $n = 1{,}000$ is 0.03071 nats. Thus

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx -0.01 + 0.03071 = 0.02071 \text{ nats.} \tag{C.1}$$

The positive centre explains why almost every rejection indicates the wrong ordering. Two-sided power retains every rejection; these counts show why its interpretation also requires the estimated direction.

C.4 Holm adjustment for the application

Table C.7 gives the raw and adjusted p-values for all six comparisons used to select the online-shopping application. Each tests equal predictor–outcome MI between two groups. Adult compares female and male records using education (16 categories) or occupation (14 categories) and income above US$50,000 (Becker and Kohavi, 1996). Its training and test partitions are combined without survey weights; records with missing occupation are excluded only from that comparison. Student Performance compares schools GP and MS using Portuguese-language course records, study time (four levels) or maternal education (five levels), and a final grade of at least 10 (Cortez, 2008). The shopping comparisons use month or weekend and purchasing among new and returning visitors.

Table C.7: Raw and Holm-adjusted p-values for the six exploratory comparisons.

	Normal Wald		Expanded Welch	
Comparison	Raw	Adjusted	Raw	Adjusted
Adult: occupation	1.18×10^{-44}	7.10×10^{-44}	1.64×10^{-42}	9.81×10^{-42}
Adult: education	1.89×10^{-34}	9.45×10^{-34}	4.93×10^{-33}	2.46×10^{-32}
Online Shoppers: month	4.71×10^{-5}	1.89×10^{-4}	3.73×10^{-4}	1.49×10^{-3}
Student Performance: study time	0.1982	0.5946	0.2339	0.7016
Student Performance: maternal education	0.8361	1	0.8468	1
Online Shoppers: weekend	0.9554	1	0.9601	1

Apply Holm's adjustment separately to each method's six p-values (Holm, 1979). Order the raw p-values and multiply them by 6, 5, 4, 3, 2 and 1. Moving through that order, replace any value below the preceding adjusted value by that value, and cap the results at one.

Bibliography

Evan Archer, Il Memming Park, and Jonathan W. Pillow. Bayesian and quasi-Bayesian estimators for mutual information from discrete data. *Entropy*, 15(5): 1738–1755, 2013. doi: 10.3390/e15051738.

Moulinath Banerjee. Likelihood ratio tests under local alternatives in regular semiparametric models. *Statistica Sinica*, 15(3):635–644, 2005. URL https://www3.stat.sinica.edu.tw/statistica/J15N3/J15N33/J15N33.html.

Barry Becker and Ronny Kohavi. Adult. UCI Machine Learning Repository, 1996. URL https://doi.org/10.24432/C5XW20. Dataset.

Robert M. Bell and Daniel F. McCaffrey. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28(2):169–181, 2002. URL https://www150.statcan.gc.ca/n1/pub/12-001-x/2002002/article/9058-eng.pdf.

Cristiano Bocci, Enrico Carlini, and Fabio Rapallo. Geometry of diagonal-effect models for contingency tables. In Marlos A. G. Viana and Henry P. Wynn, editors, *Algebraic Methods in Statistics and Probability II*, volume 516 of *Contemporary Mathematics*, pages 61–73. American Mathematical Society, Providence, Rhode Island, 2010. doi: 10.1090/conm/516/10166. URL https://arxiv.org/abs/0908.0232.

Terry Bossomaier, Lionel Barnett, Michael Harré, and Joseph T. Lizier. *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer, Cham, 2016. doi: 10.1007/978-3-319-43222-9.

Richard P. Brent. *Algorithms for Minimization without Derivatives*. Prentice-Hall, Englewood Cliffs, New Jersey, 1973.

David R. Brillinger. Some data analyses using mutual information. *Brazilian Journal of Probability and Statistics*, 18(2):163–182, 2004. URL https://www.stat.berkeley.edu/~brill/Papers/MIBJPS.pdf.

Anne Chao and Tsung-Jen Shen. Nonparametric estimation of Shannon's index of diversity when there are unseen species in sample. *Environmental and Ecological Statistics*, 10(4):429–443, 2003. doi: 10.1023/A:1026096204727.

EunYi Chung and Joseph P. Romano. Exact and asymptotically robust permutation tests. *Annals of Statistics*, 41(2):484–507, 2013. doi: 10.1214/13-AOS1090.

Paulo Cortez. Student Performance. UCI Machine Learning Repository, 2008. URL https://doi.org/10.24432/C5TG7T. Dataset.

Robert Drake and Apratim Guha. A mutual information-based k-sample test for discrete distributions. *Journal of Applied Statistics*, 41(9):2011–2027, 2014. doi: 10.1080/02664763.2014.899325.

Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.

Michael P. Fay and Barry I. Graubard. Small-sample adjustments for Wald-type tests using sandwich estimators. *Biometrics*, 57(4):1198–1206, 2001. doi: 10.1111/j.0006-341X.2001.01198.x.

Apratim Guha and Tom Chothia. A two sample test based on mutual information. *Calcutta Statistical Association Bulletin*, 66(1–2):39–54, 2014. doi: 10.1177/0008068320140103.

Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974. doi: 10.1080/01621459.1974.10482962.

Damián G. Hernández and Inés Samengo. Estimating the mutual information between two discrete, asymmetric variables with limited samples. *Entropy*, 21(6):623, 2019. doi: 10.3390/e21060623.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL https://www.jstor.org/stable/4615733.

Kermit Hutcheson. A test for comparing diversities based on the Shannon formula. *Journal of Theoretical Biology*, 29(1):151–154, 1970. doi: 10.1016/0022-5193(70)90124-4.

Marcus Hutter. Distribution of mutual information. In *Advances in Neural Information Processing Systems 14*, pages 399–406, 2002. Conference held in 2001; arXiv:cs/0112019.

Marcus Hutter and Marco Zaffalon. Distribution of mutual information from complete and incomplete data. *Computational Statistics & Data Analysis*, 48(3): 633–657, 2005. doi: 10.1016/j.csda.2004.03.010.

Guido W. Imbens and Michal Kolesár. Robust standard errors in small samples: Some practical advice. *The Review of Economics and Statistics*, 98(4):701–712, 2016. doi: 10.1162/REST_a_00552.

Arnold Janssen. Studentized permutation tests for non-i.i.d. hypotheses and the generalized Behrens–Fisher problem. *Statistics & Probability Letters*, 36(1):9–21, 1997. doi: 10.1016/S0167-7152(97)00043-6.

Kirthevasan Kandasamy, Akshay Krishnamurthy, Barnabas Poczos, Larry Wasserman, and James M. Robins. Nonparametric von Mises estimators for entropies, divergences and mutual informations. In *Advances in Neural Information Processing Systems 28*, pages 397–405, 2015.

Göran Kauermann and Raymond J. Carroll. A note on the efficiency of sandwich covariance matrix estimation. *Journal of the American Statistical Association*, 96 (456):1387–1396, 2001. doi: 10.1198/016214501753382309.

Marius Marinescu and Costel Balcau. On the use of mutual information for testing independence, 2025. arXiv:2502.17636.

George A. Miller. Note on the bias of information estimates. In Henry Quastler, editor, *Information Theory in Psychology: Problems and Methods*, pages 95–100. Free Press, Glencoe, Illinois, 1955.

Roelof Moddemeijer. On estimation of entropy and mutual information of continuous distributions. *Signal Processing*, 16(3):233–248, 1989. doi: 10.1016/0165-1684(89)90132-1.

Roelof Moddemeijer. A statistic to estimate the variance of the histogram-based mutual information estimator based on dependent pairs of observations. *Signal Processing*, 75(1):51–63, 1999. doi: 10.1016/S0165-1684(98)00224-2.

Ricardo Mora and Javier Ruiz-Castillo. The statistical properties of the mutual information index of multigroup segregation. Working Paper, Economic Series (48) 09-84, Universidad Carlos III de Madrid, Departamento de Economía, 2009.

Ricardo Mora and Javier Ruiz-Castillo. Entropy-based segregation indices. *Sociological Methodology*, 41(1):159–194, 2011. doi: 10.1111/j.1467-9531.2011.01237.x.

Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019. doi: 10.1002/sim.8086.

Ilya Nemenman, Fariel Shafee, and William Bialek. Entropy and inference, revisited. In Thomas G. Dietterich, Suzanna Becker, and Zoubin Ghahramani, editors, *Advances in Neural Information Processing Systems*, volume 14, pages 471–478. MIT Press, 2002. URL https://papers.neurips.cc/paper_files/paper/2001/file/d46e1fcf4c07ce4a69ee07e4134bcef1-Paper.pdf.

Wei Pan and Melanie M. Wall. Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*, 21 (10):1429–1441, 2002. doi: 10.1002/sim.1142.

Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003. doi: 10.1162/089976603321780272.

Stefano Panzeri and Alessandro Treves. Analytical estimates of limited sampling biases in different information measures. *Network: Computation in Neural Systems*, 7(1):87–107, 1996. doi: 10.1088/0954-898X/7/1/006.

Angelo Piga, Lluc Font-Pomarol, Marta Sales-Pardo, and Roger Guimerà. Bayesian estimation of information-theoretic metrics for sparsely sampled distributions. *Chaos, Solitons & Fractals*, 180:114564, 2024. doi: 10.1016/j.chaos.2024.114564.

Mark S. Roulston. Estimating the errors on measured entropy and mutual information. *Physica D: Nonlinear Phenomena*, 125(3–4):285–294, 1999. doi: 10.1016/S0167-2789(98)00269-3.

C. Sakar and Yomi Kastro. Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository, 2018. URL https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset. Dataset.

Franklin E. Satterthwaite. An approximate distribution of estimates of variance components. *Biometrics Bulletin*, 2(6):110–114, 1946. doi: 10.2307/3002019.

Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948. Parts I and II.

Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.

Abraham Wald. Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, 54(3):426–482, 1943. doi: 10.1090/S0002-9947-1943-0012401-3.

Bernard L. Welch. The generalization of "Student's" problem when several different population variances are involved. *Biometrika*, 34(1–2):28–35, 1947. doi: 10.1093/biomet/34.1-2.28.

Samuel S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, 1938. doi: 10.1214/aoms/1177732360.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.